

ESE 5700

FINAL PROJECT REPORT

CONFIGURABLE LOGIC BLOCK (CLB)

Table of Contents

Introduction.....	4
Proposed Design	4
Figure 1: Proposed Design	4
6T SRAM.....	5
Proposed Design	5
Figure 2: Proposed Design for 6TSRAM	5
Figure 3: Transistor Sizing for 6TSRAM	6
Schematic.....	7
Figure 4: 6TSRAM Schematic	7
Symbol.....	7
Figure 5: 6TSRAM Symbol	7
Functionality Test.....	8
Figure 6: Test Schematic for 6TSRAM	8
Figure 7: Verification Result for 6TSRAM	9
6TSRAM Array.....	9
Schematic.....	9
Figure 8: 6TSRAM Array.....	9
Symbol.....	10
Figure 9: 6TSRAM Array Symbol	10
Verification.....	10
Figure 10: Test Schematic for 6TSRAM Array	10
Figure 11: Verification Result for 6TSRAM Array	11
Non-Overlapping Clock Generator.....	11
Figure 12: Proposed Non-Overlapping Clock Design using NAND Gates	11
Schematic.....	12
Figure 13: NAND Schematic and Symbol for Clock	12
Figure 14: Clock Schematic	12
Symbol.....	13
Figure 15: Clock Symbol.....	13
Verification.....	13
Figure 16: Test Schematic for Clock	13
Figure 17: Verification Result for Clock.....	13

D-Flip Flop	14
Schematic.....	14
Figure 18: DFF Schematic.....	14
Symbol.....	14
Figure 19: DFF Symbol.....	14
Verification.....	15
Figure 20: Test Schematic for DFF	15
Figure 21: Verification Result for DFF	15
SIPO	16
Schematic.....	16
Figure 22: SIPO Schematic	16
Symbol.....	16
Figure 23: SIPO Symbol	16
Verification.....	17
Figure 24: Test Schematic for SIPO	17
Figure 25: Verification Result for SIPO	17
MUX.....	18
Schematic.....	18
Figure 26: MUX Schematic.....	18
Symbol.....	19
Figure 27: MUX Symbol.....	19
Verification.....	19
Figure 28: Test Schematic for LUT	19
Figure 29: Verification Result for MUX.....	20
LUT.....	20
Schematic.....	20
Figure 30: 16X1 LUT Schematic	20
LUT Symbol	21
Figure 31: 16X1 LUT Symbol.....	21
Verification.....	21
Figure 32: Test Schematic for 16X1 LUT	21
Figure 33: Verification Result for 16X1 LUT	22
LOAD LOGIC	22
Schematic.....	22
Figure 34: Load Logic Schematic	22
Symbol.....	23
Figure 35: Load Logic Symbol.....	23
SIPO-SRAM Verification	23

Figure 36: SIPO - SRAM Test Schematic	23
Figure 37: Verification Result for SIPO – SRAM	23
SIPO-SRAM-LUT Verification.....	24
Figure 38: SIPO-SRAM-LUT Schematic.....	24
Figure 39: Verification Result for SIPO - SRAM – LUT	25
CLB.....	26
Schematic	26
Figure 40: CLB schematic.....	26
Symbol.....	26
Figure 41: CLB Symbol	26
Verification.....	27
Figure 42: CLB Test Schematic.....	27
Figure 43: Verification Result for CLB	27
Optimisation Steps Taken	28
FOM.....	29
Frequency.....	29
Figure 44: Max Frequency giving correct output at 1.66ns time period.....	29
Figure 45: Incorrect Output at 1.65ns time period.....	29
Average Energy	30
Loading Energy	30
Figure 46: Loading Energy from 0 to 28ns.....	30
Figure 47: Loading Energy Calculation	30
Active Energy.....	31
Figure 48: Active Energy from 28ns to 54ns	31
Figure 49: Active Energy Calculation	31
Area.....	32
Cell Area	32
2X1 Mux Area.....	32
Summary Table	32

Introduction

The increasing complexity of digital systems has driven the demand for highly efficient and versatile Configurable Logic Blocks (CLBs), which form the backbone of modern Field Programmable Gate Arrays (FPGAs). This project focuses on the design and simulation of a CLB using Cadence, integrating key components such as a -

- 1x16 Serial-In Parallel-Out (SIPO) shift register for sequential data input
- Parallel data transfer to 16 6T SRAM memory cells
- 16x1 Look-Up Table (LUT) implemented using multiplexers (MUX) and
- D flip-flop for synchronous state storage and output generation
- A clock generator

The CLB is designed to perform configurable logic operations with high speed and flexibility. The system begins with a 1x16 SIPO shift register, responsible for serially receiving input data and converting it into a parallel format. This parallel data is then stored in an array of 16 6T SRAM cells, enabling efficient and robust memory operations. The stored data is subsequently fed into a 16x1 LUT, which uses multiplexers to provide a programmable logic function based on predefined truth tables. Finally, the LUT output is temporarily stored in a D flip-flop to ensure synchronized data availability, enabling consistent and stable outputs.

This project leverages Cadence tools to design, simulate, and verify the functionality of each component within the CLB.

Proposed Design

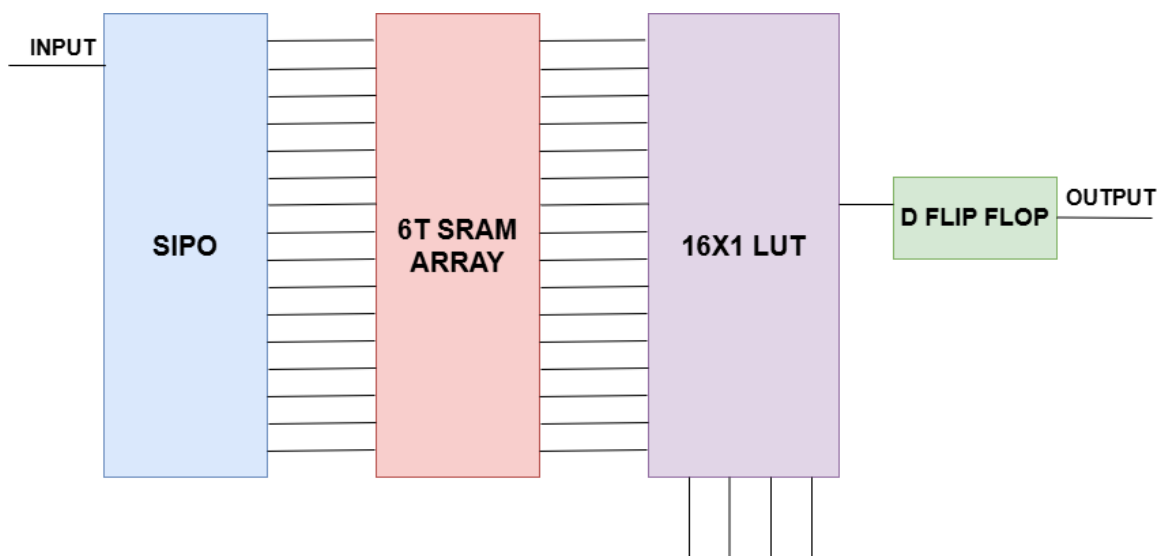


Figure 1: Proposed Design

6T SRAM

A 6-Transistor (6T) SRAM cell is a fundamental memory storage unit in digital integrated circuits, consisting of six transistors arranged to create a bistable circuit capable of storing one bit of data.

Key Components

- Two cross-coupled inverters (4 transistors)
- Two access transistors
- Bit lines for data transfer
- Word line for cell selection

Operation Modes

1. Write Operation:
 - Data driven onto bit lines BL and BL'
 - Raise Word line (WL)
2. Read Operation:
 - Pre-charge bit lines BL and BL'
 - Raise Word line (WL)

Proposed Design

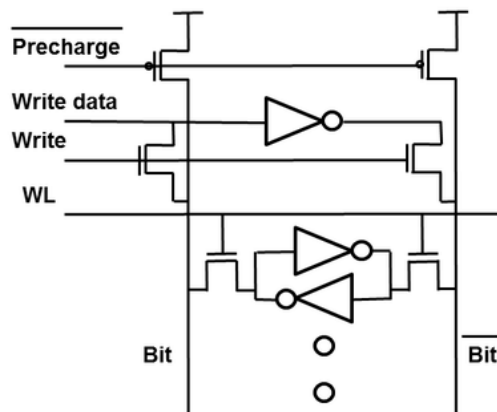


Figure 2: Proposed Design for 6TSRAM

We have also incorporated the pre-charge and write enable circuit to the 6TSRAM.

The Pre-charge circuit –

- Consists of PMOS transistors and gets active when both Write and WL are **LOW**.
- Pulls bit lines to a predefined voltage (usually VDD)
- Occurs before each read cycle

The Write Enable circuit –

- Consists of 2 NMOS pass transistors and an inverter.
- Controls data writing into SRAM cell and manages input data path.
- Gets active only when the Write is **HIGH**

In a standard 6T-SRAM, for proper write and read operation to take place, we need to make sure that the Cell Ratio (CR) and Pull-up Ratio (PR) is maintained.

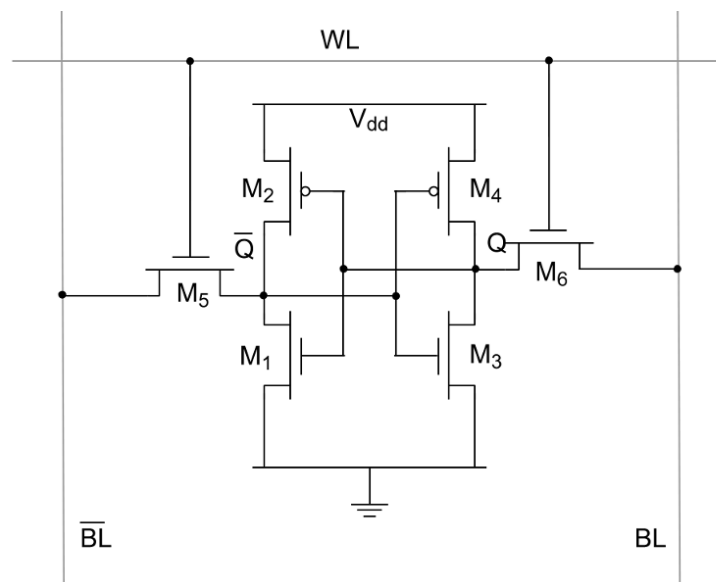


Figure 3: Transistor Sizing for 6TSRAM

$$CR = \frac{W1/L1}{W5/L5}$$

$$PR = \frac{W4/L4}{W6/L6}$$

The two basic requirements which dictate the (W/L) ratios are:

- The data-read operation should not destroy the stored information in the SRAM cell
- The cell should allow modification of the stored information during the data-write phase.

For an uncorrupted read operation, $M1 > M5$ in the 6T SRAM above.

For a proper write operation, $M6 \gg M4$ in the 6T SRAM above.

Hence, we resized our circuit accordingly as can be seen from the image below (*Figure 3*)

Pull Down NMOS – 180nm

Write Enable NMOS – 240nm

Rest all works well with minimum size – 120nm

Schematic

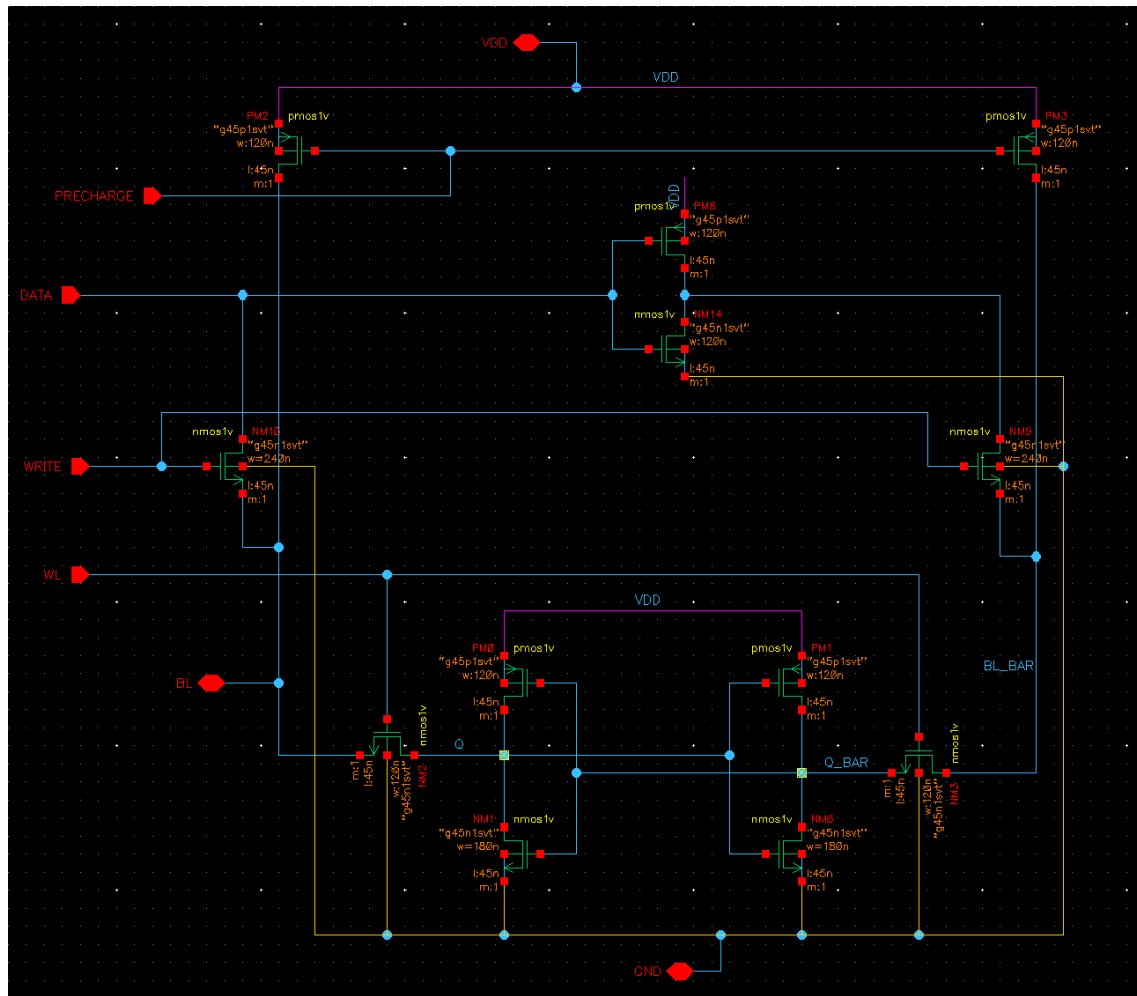


Figure 4: 6T1SRAM Schematic

Symbol

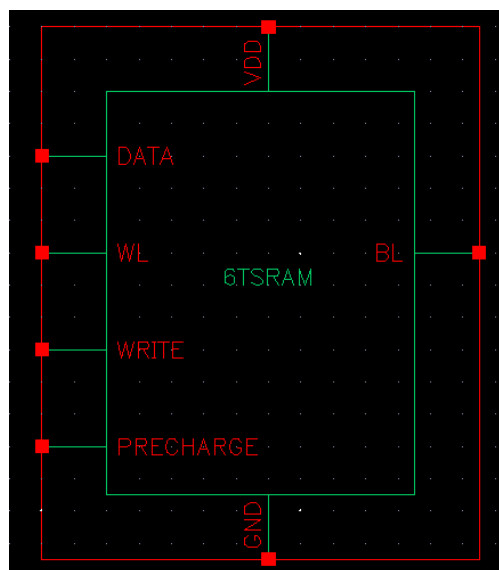


Figure 5: 6T1SRAM Symbol

Functionality Test

In the test schematic below (Fig. 6), an OR gate is used with inputs from the Word Line (WL) and Write Enable signals to control the Pre-charge logic. Since the Pre-charge circuit is active LOW due to the use of PMOS transistors, the OR gate ensures that the Pre-charge is activated only when both the WL and Write Enable signals are LOW. This means that the Pre-charge will only occur during the read operation.

In Fig. 4, which illustrates the 6T SRAM schematic, the Write Enable circuit incorporates an NMOS pass transistor. During a write operation, this NMOS transistor becomes active, allowing data to flow through it into the memory cell. However, it is well-known that an NMOS transistor passes a strong 0 but a weak 1, resulting in a threshold voltage drop (V_{th}) when writing a logic 1 to the memory.

To address this issue and ensure accurate read operations, a buffer is added before reading the bit line (BL). This buffer compensates for the threshold voltage drop, restoring the logic level to its intended state and enhancing the reliability of the read operation.

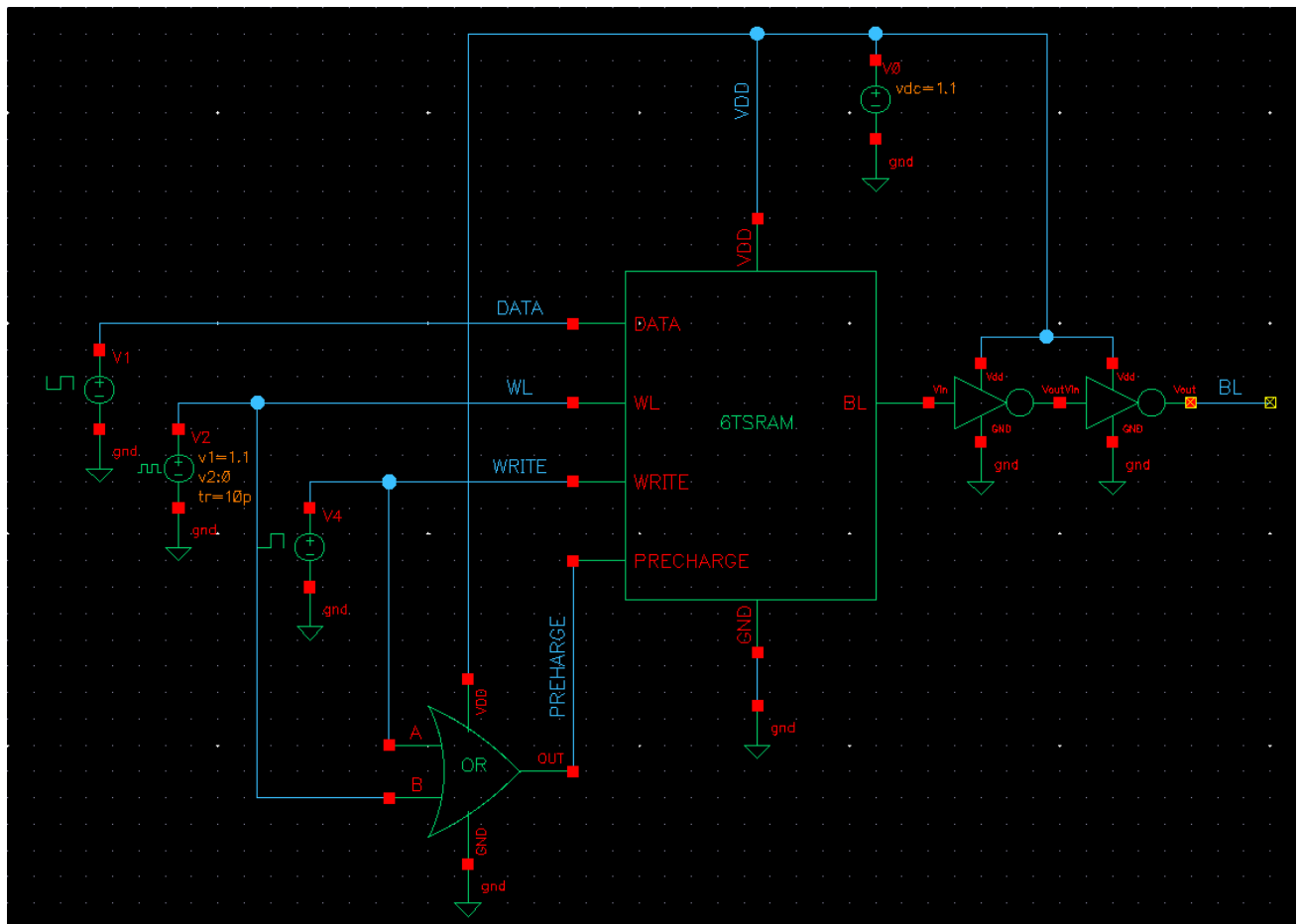


Figure 6: Test Schematic for 6TSRAM

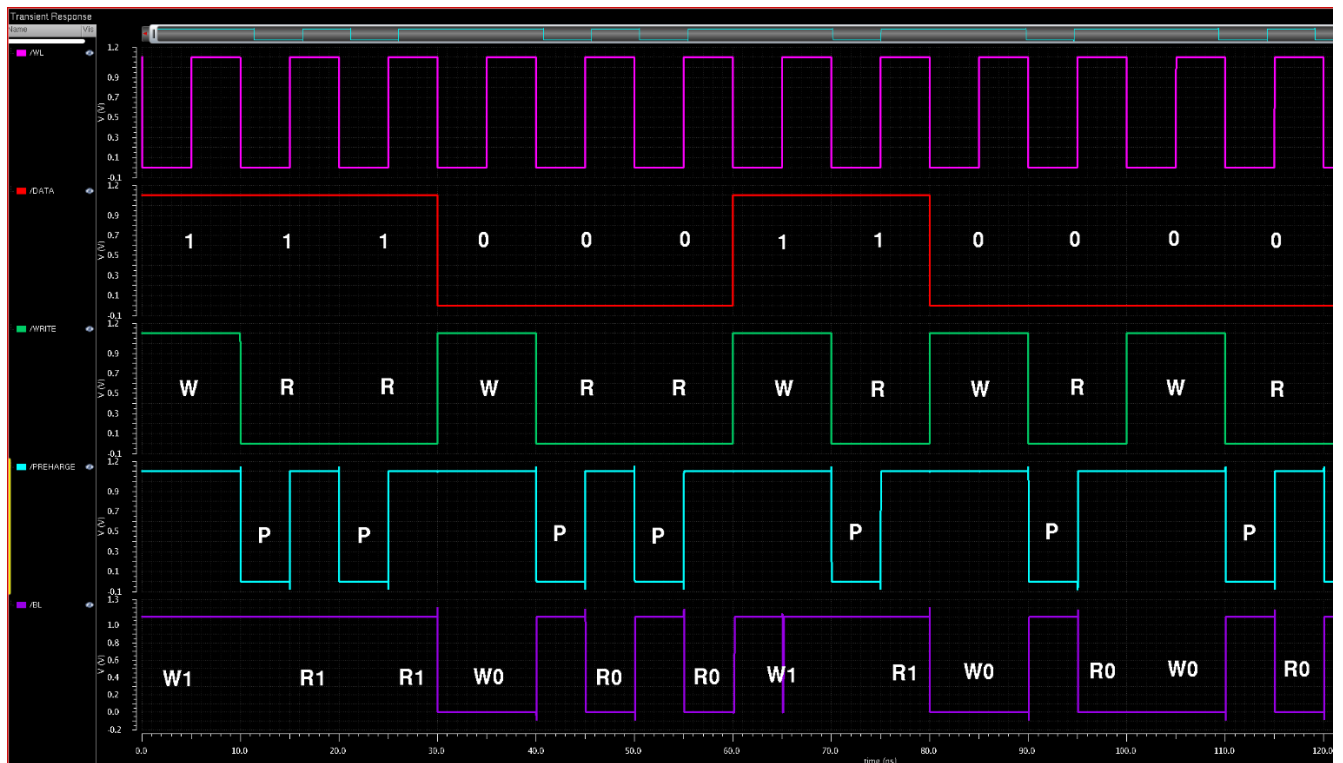


Figure 7: Verification Result for 6TSRAM

To ensure that the data is being stored correctly in the memory, we first write data into the cell and then read it for 2 consecutive cycles to make sure it writes and holds the data correctly. The same can also be seen from the graph obtained in Fig. 7. We have followed the below pattern for the test and received the expected output –

W1R1R1 W0R0R0 W1R1 W0R0 W0R0

6TSRAM Array

Schematic

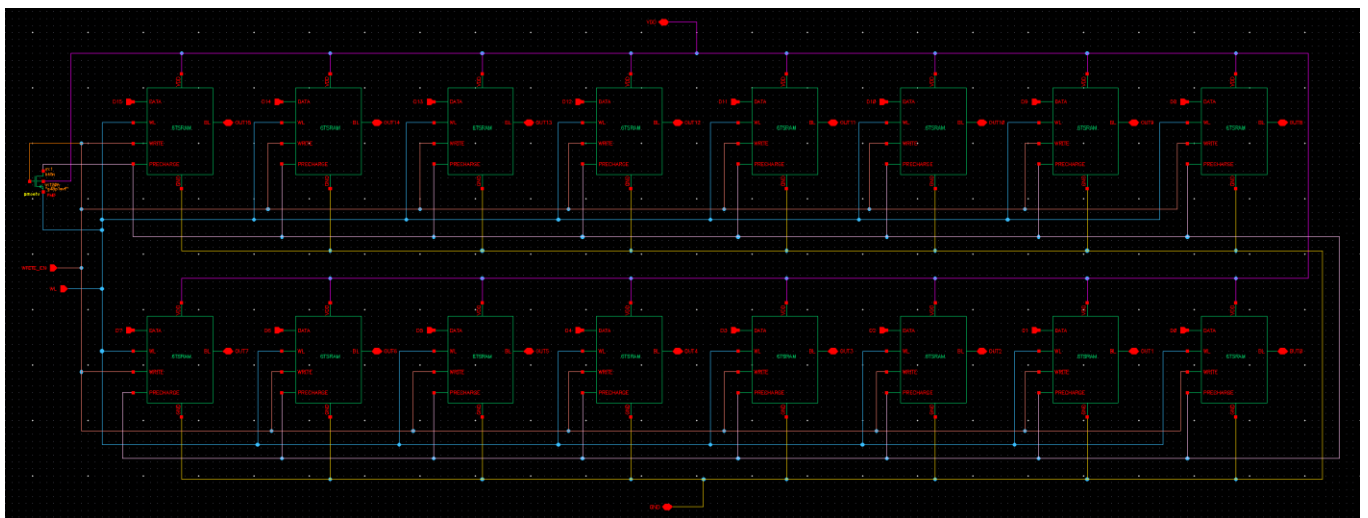


Figure 8: 6TSRAM Array

Symbol

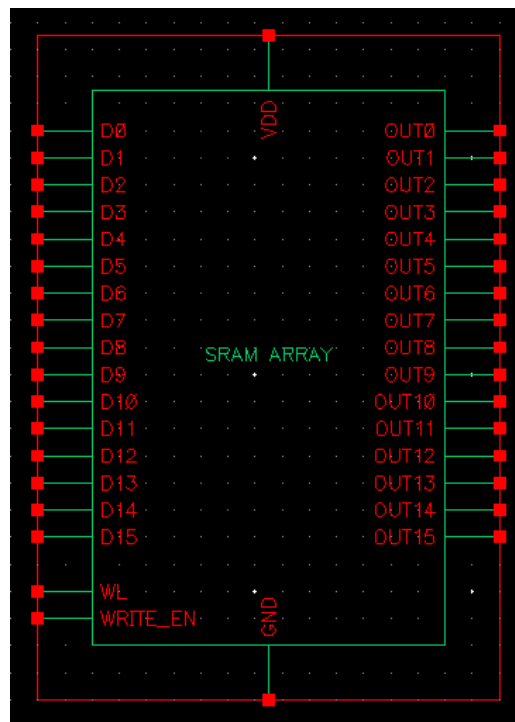


Figure 9: 6TSRAM Array Symbol

Verification

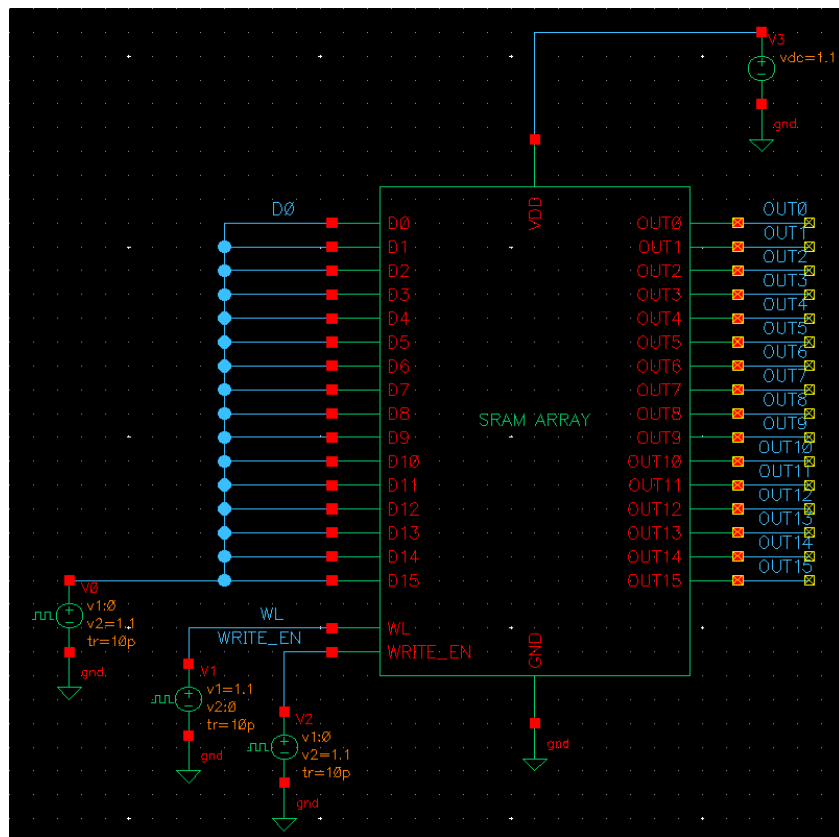


Figure 10: Test Schematic for 6TSRAM Array

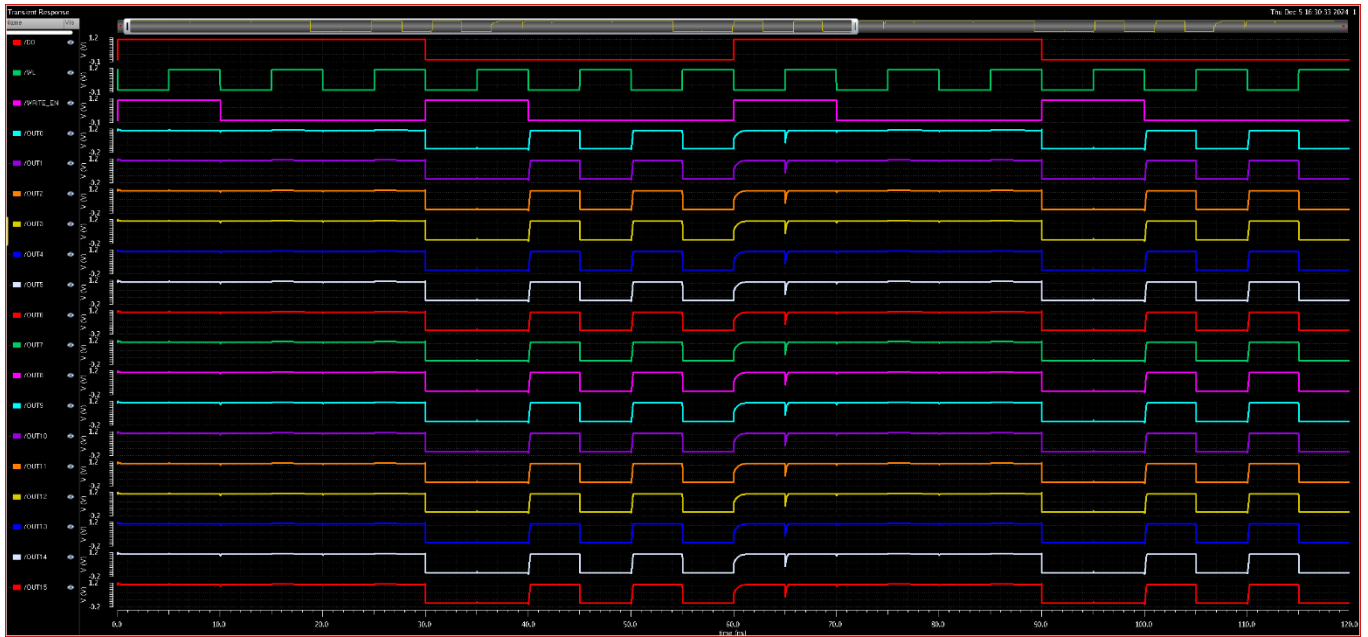


Figure 11: Verification Result for 6TSRAM Array

All 16 SRAMS in the SRAM Array are reading and writing the data correctly.

Non-Overlapping Clock Generator

A non-overlapping clock generates two complementary clock signals (ϕ_1 and ϕ_2). It is needed in this CLB design to ensure proper timing and synchronization between the different circuit components.

The key reasons are:

Sequential data transfer: The 1x16 SIPO register requires a non-overlapping clock to shift data in serially and transfer it in parallel to the 16 6TSRAM cells.

Memory access coordination: The non-overlapping clock ensures that read and write operations to the 6TSRAM cells are properly timed and do not interfere with each other.

Latch-based logic: Much needed in the D flip-flop to reliably capture and hold the output data.

Avoiding glitches: They also help prevent potential logic glitches that could occur if clock signals were to overlap and cause momentary contention.

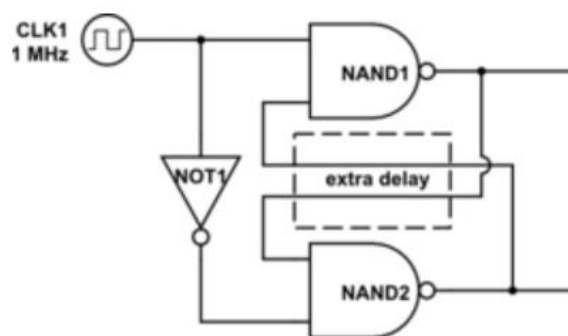


Figure 12: Proposed Non-Overlapping Clock Design using NAND Gates

Schematic

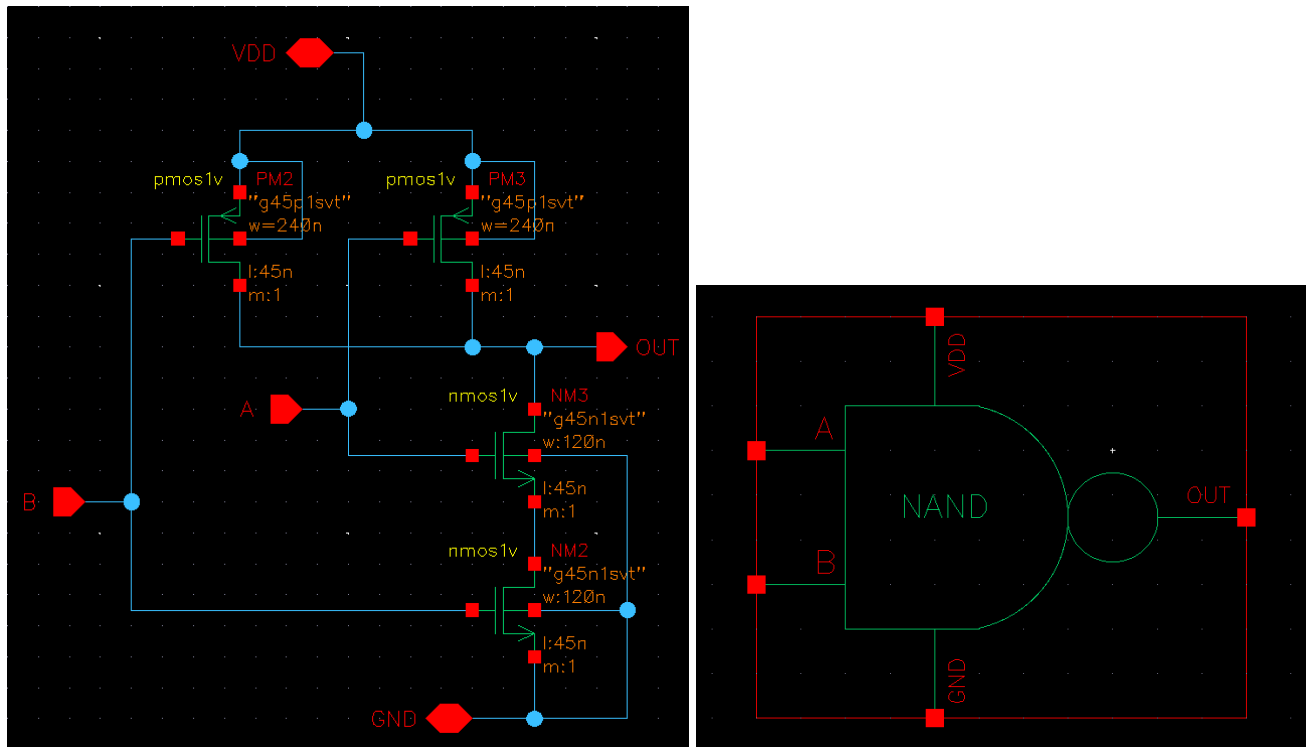


Figure 13: NAND Schematic and Symbol for Clock

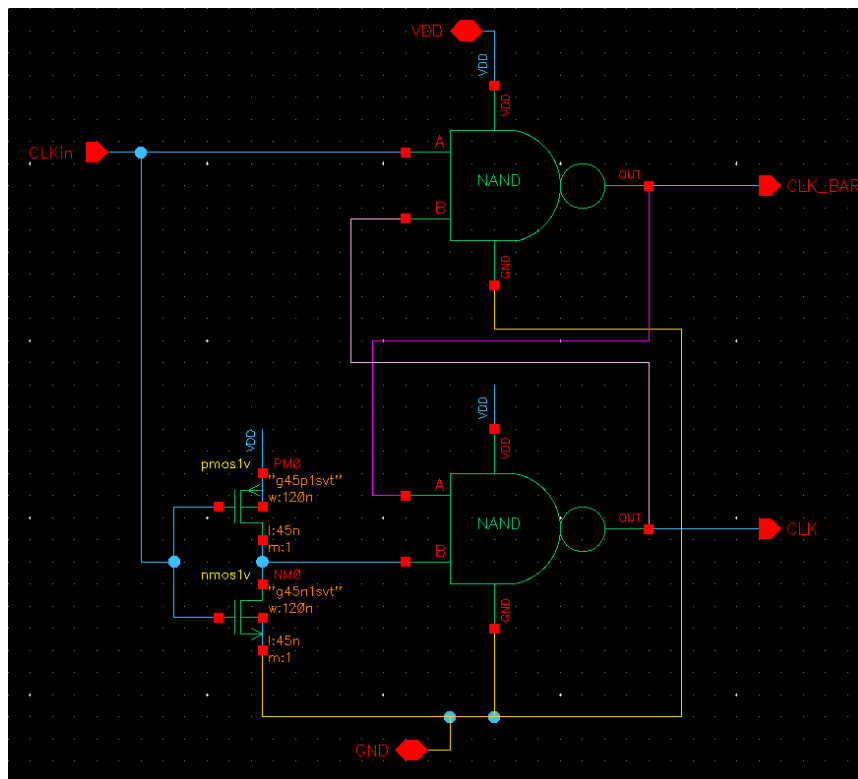


Figure 14: Clock Schematic

Symbol

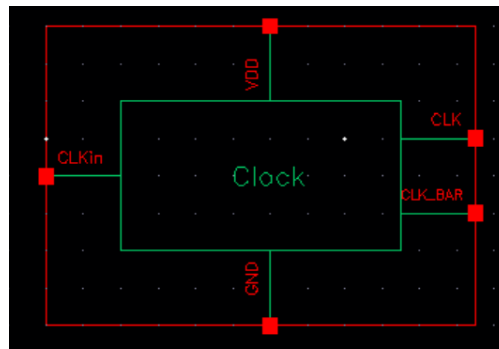


Figure 15: Clock Symbol

Verification

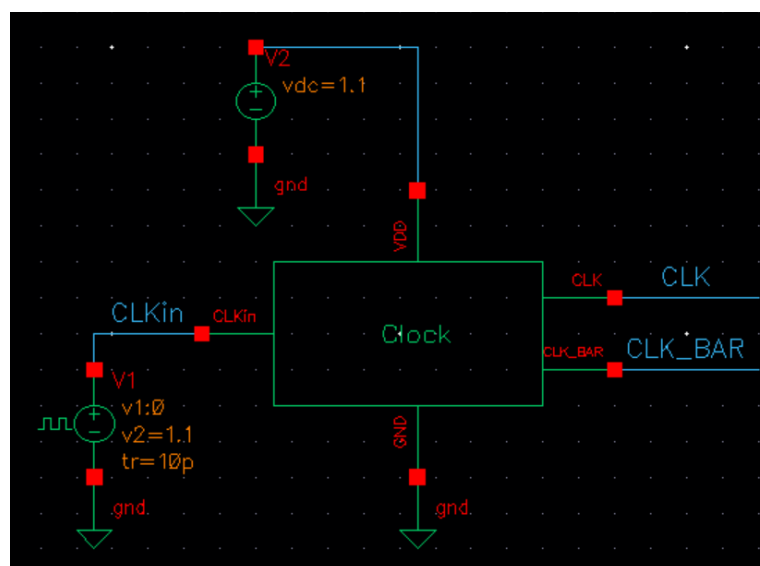


Figure 16: Test Schematic for Clock

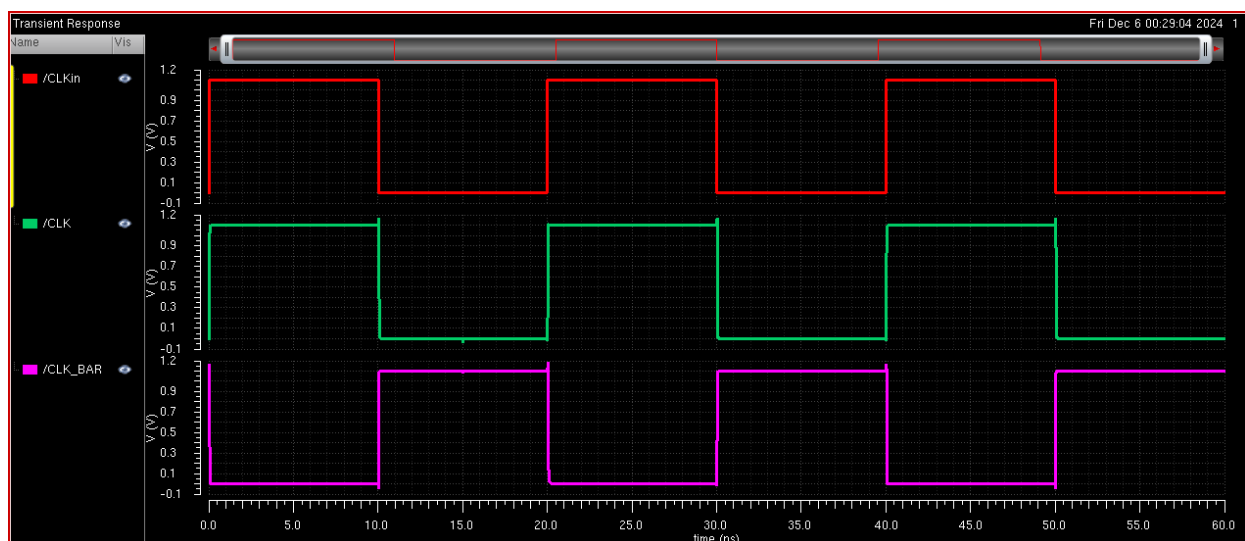


Figure 17: Verification Result for Clock

From the graph we can see that we are getting two signals – CLK and CLK_BAR which are completely out of phase. and are non-overlapping. We will be using these out of phase signals to synchronize other components.

D-Flip Flop

The D flip-flop is a critical sequential element in the Configurable Logic Block (CLB) design, serving two key roles:

- **In the SIPO Register:**
D flip-flops are used to sequentially store and shift the incoming serial data.
- **At the End of the CLB:**
At the output stage, the D flip-flop temporarily stores the result of the logic operation performed by the LUT. It acts as a buffer, providing a stable and reliable output until the next clock cycle.

Schematic

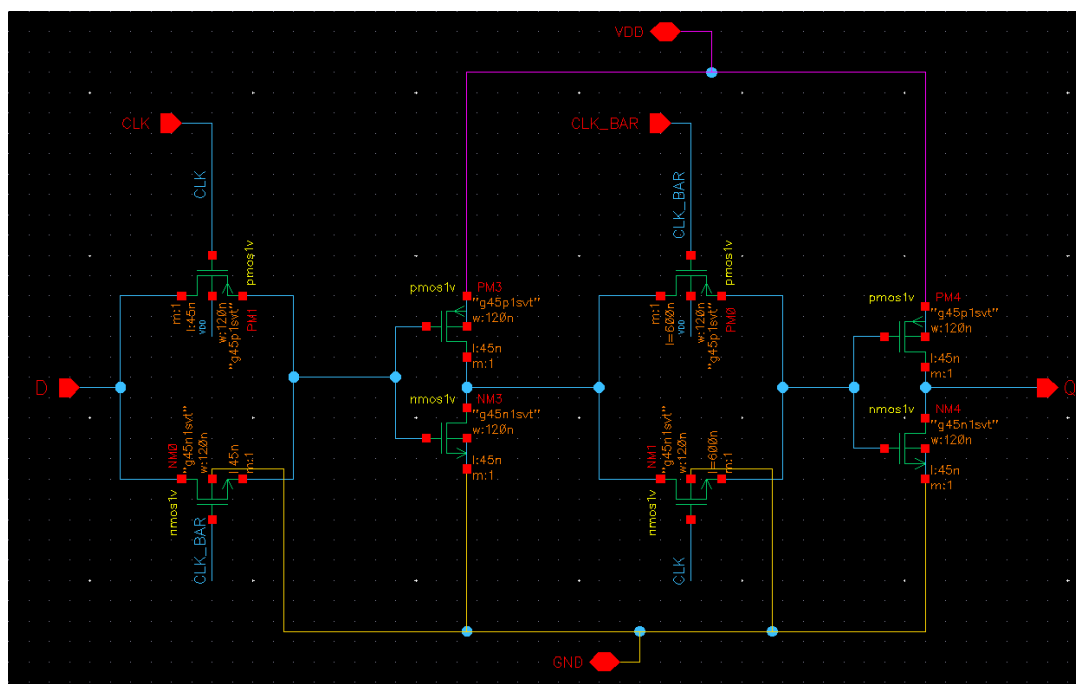


Figure 18: DFF Schematic

Symbol

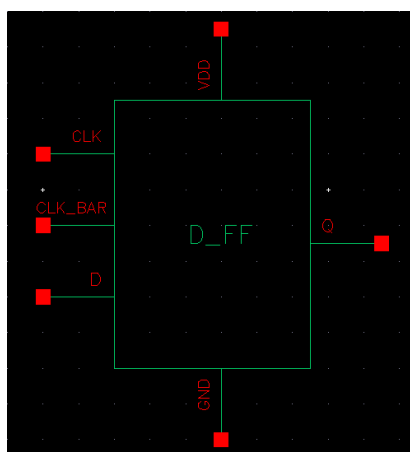


Figure 19: DFF Symbol

Verification

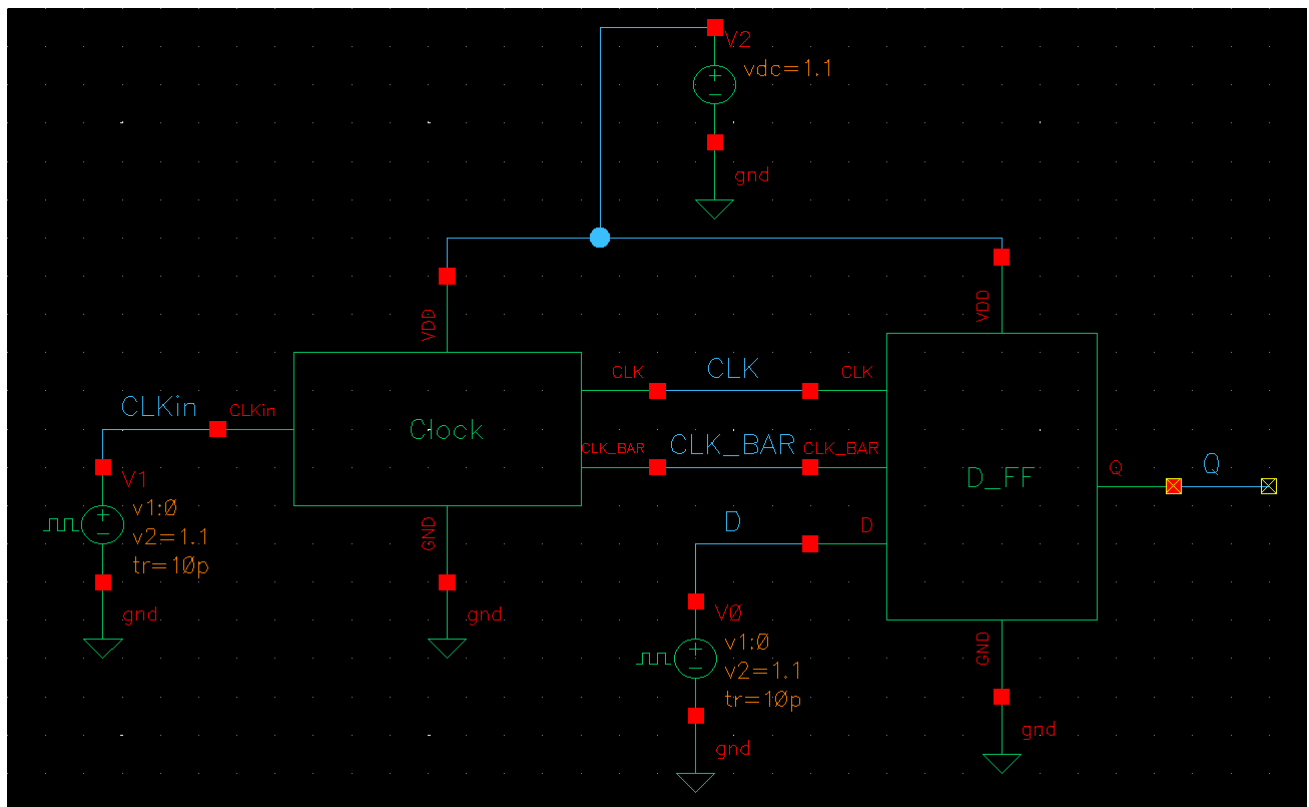


Figure 20: Test Schematic for DFF

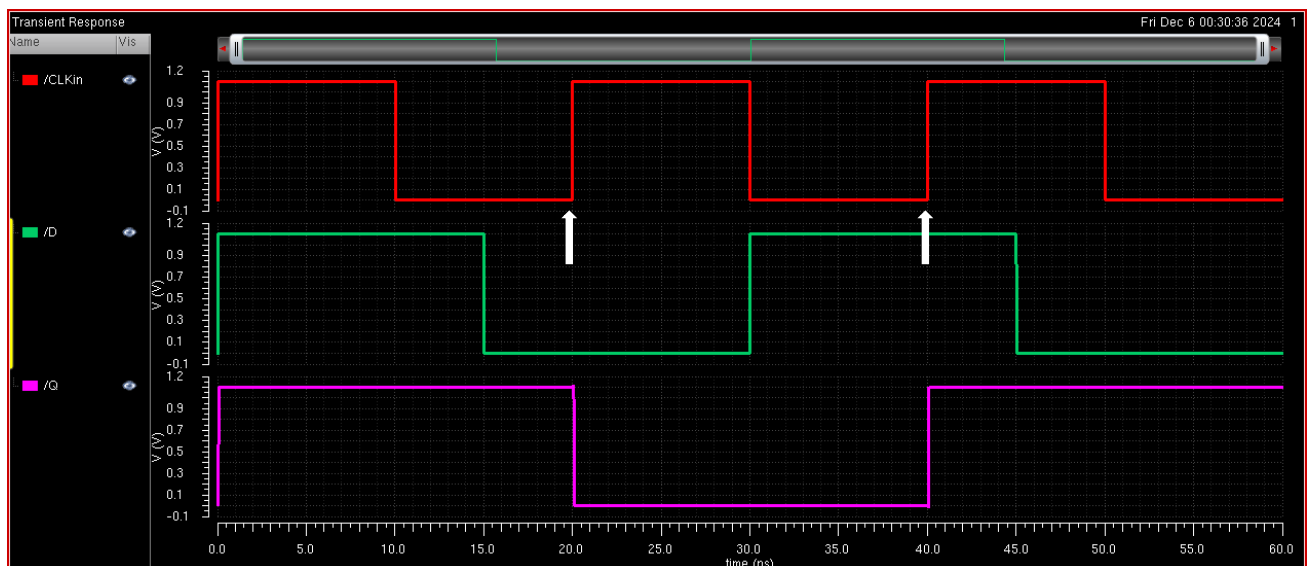


Figure 21: Verification Result for DFF

The D flip-flop in this CLB design has a delay in the CLK-Q path. This is because the data is latched at each positive clock edge and maintained until the next positive edge. This synchronizes the data flow and ensures stable output.

SIPO

The SIPO (Serial-In Parallel-Out) register is used to load the 16-bit serialized data into the SRAM array. It consists of 16 chained D flip-flops.

Key SIPO functionality:

- Converts serial data stream into parallel 16-bit format.
- Stores the input data in the flip-flops and transfers it to the next stage with each clock cycle.
- Transfers the 16-bit data in parallel to the SRAM cells.

Operation:

- Serial data is clocked into the first D flip-flop
- On each clock cycle, data shifts through the chain of 16 flip-flops
- After 16 clock cycles, the full 16-bit data is stored in the SIPO register
- The parallel 16-bit output is then transferred to the SRAM cells

Schematic

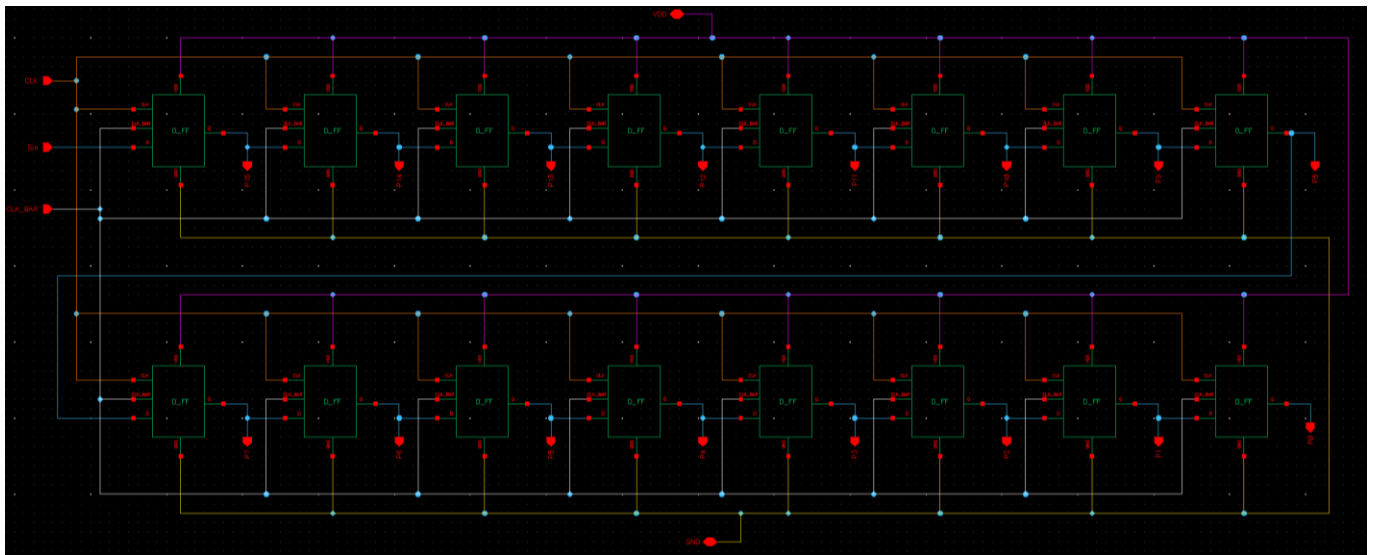


Figure 22: SIPO Schematic

Symbol

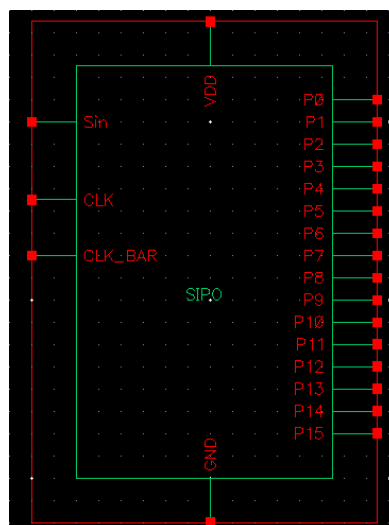


Figure 23: SIPO Symbol

Verification

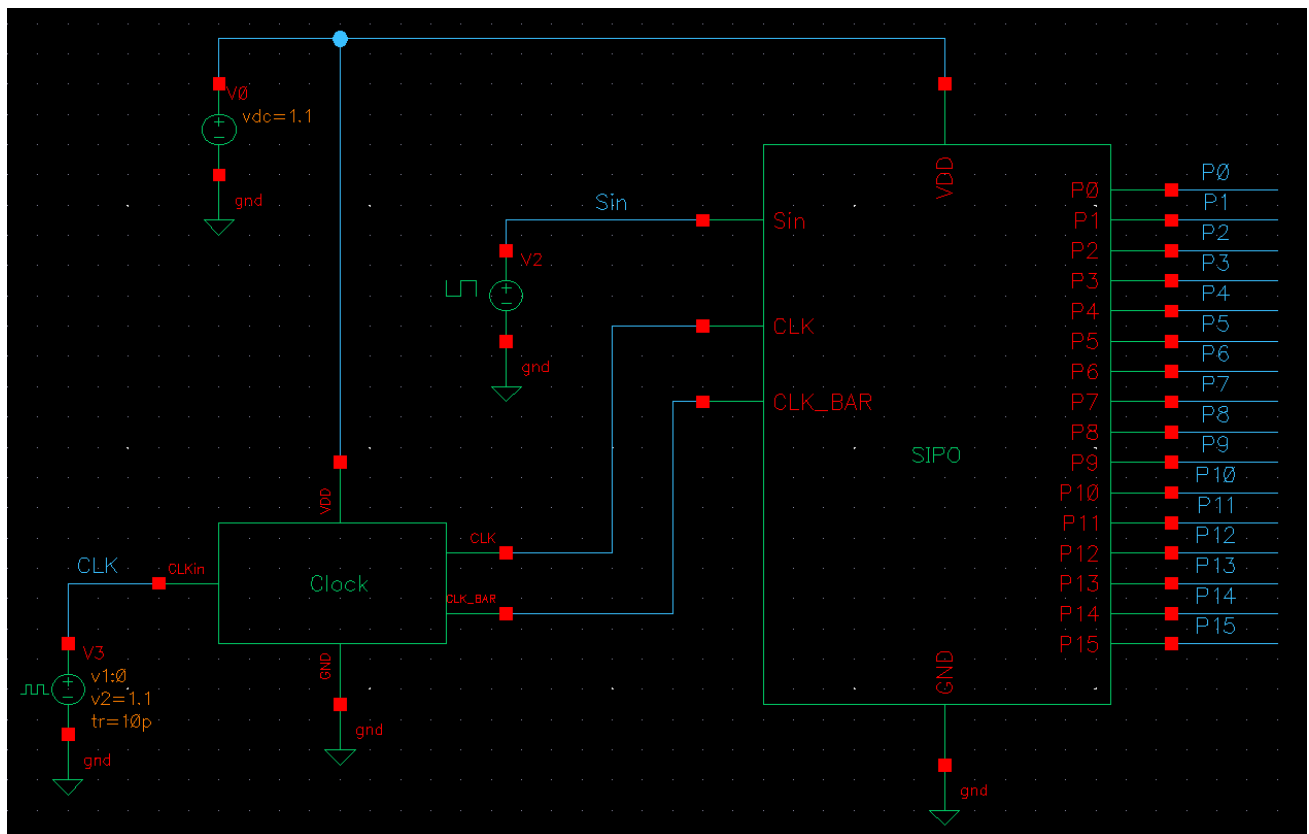


Figure 24: Test Schematic for SIPO

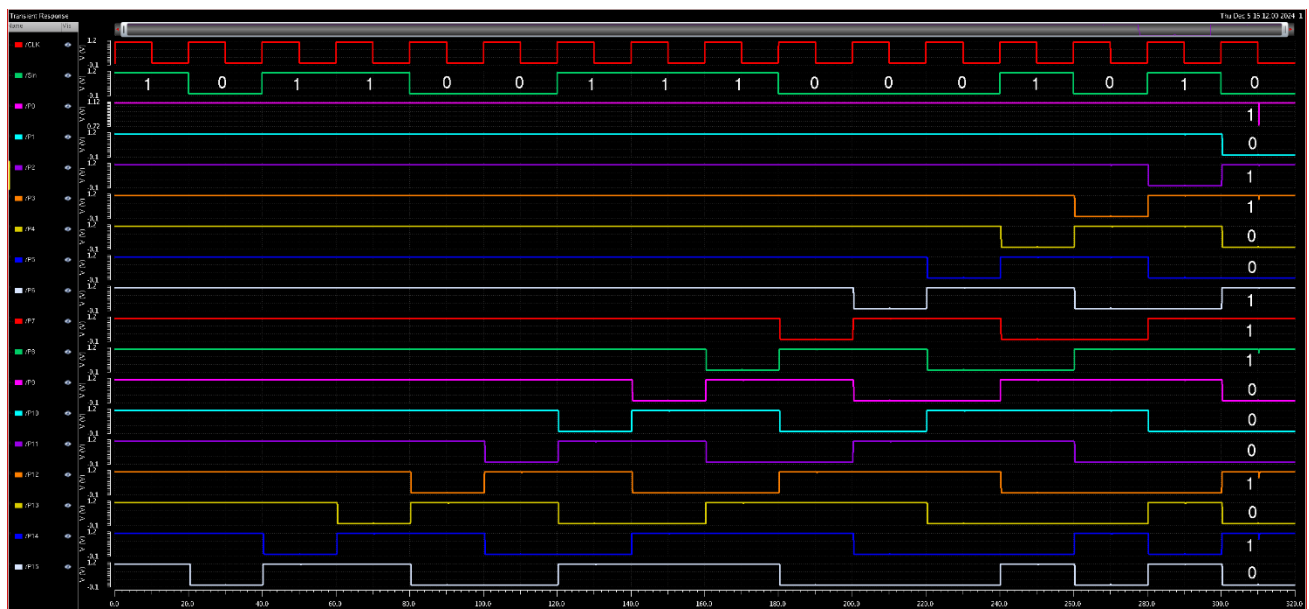


Figure 25: Verification Result for SIPO

To validate the functionality of the Serial-In Parallel-Out (SIPO) shift register, a main clock is utilized to generate two complementary clock signals: CLK and CLK_BAR (out-of-phase). The main clock operates with a 20 ns period, corresponding to a frequency of 50 MHz, ensuring precise timing for the SIPO operation.

Test Procedure:

1. **Input Data Loading:** A 16-bit binary input sequence **1011001110001010** is applied to the serial input of the SIPO.
2. **Clock Cycles:** The input data is shifted into the SIPO register sequentially, one bit per clock cycle.
3. **Parallel Output Validation:** The register takes 16 clock cycles to fully load the data, as it has 16 D flip-flops corresponding to the 16 output bits.

At the 16th clock cycle, all 16 bits of the input data are correctly reflected at the parallel output of the SIPO as can be seen from Fig. 25.

MUX

Schematic

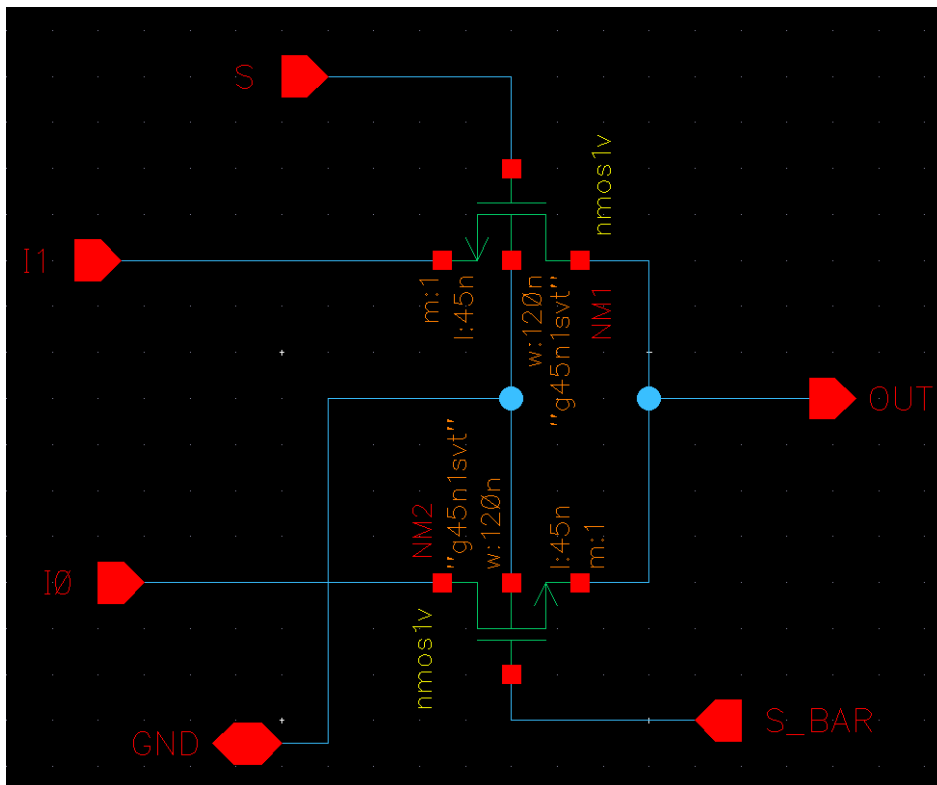


Figure 26: MUX Schematic

To optimize the 2x1 pass transistor-based MUX design for the 16x1 LUT, the inverter is moved out of individual 2x1 MUXes and each select line in the LUT is provided a common inverter. This approach saves significant area by eliminating the need for a separate inverter circuit. The S-bar value is now directly available to all 16 MUX units in the LUT.

Symbol

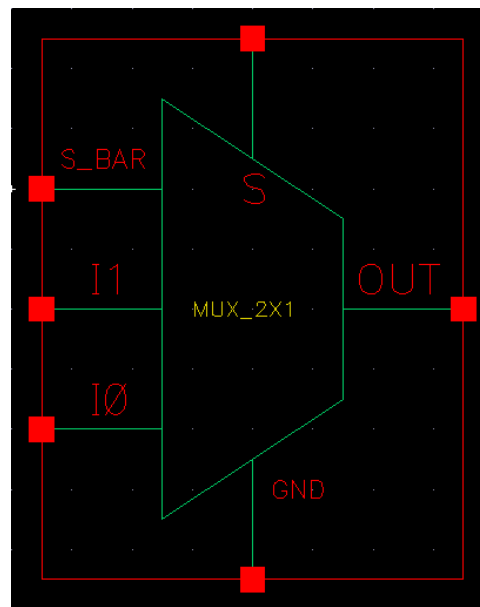


Figure 27: MUX Symbol

Verification

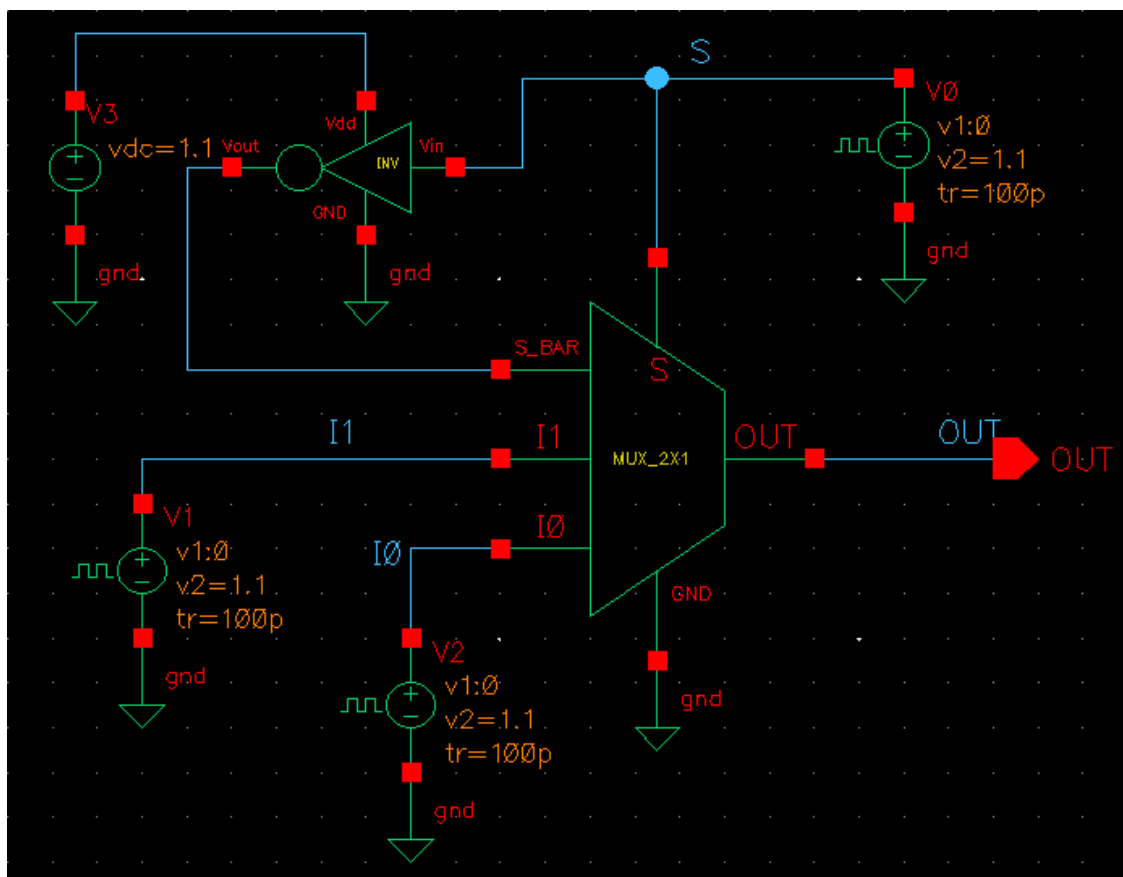


Figure 28: Test Schematic for LUT

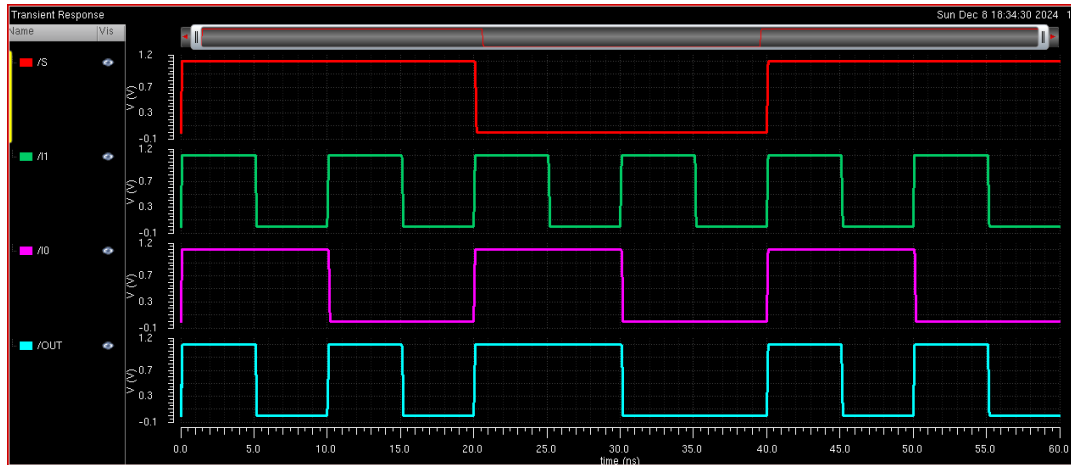


Figure 29: Verification Result for MUX

LUT

Schematic

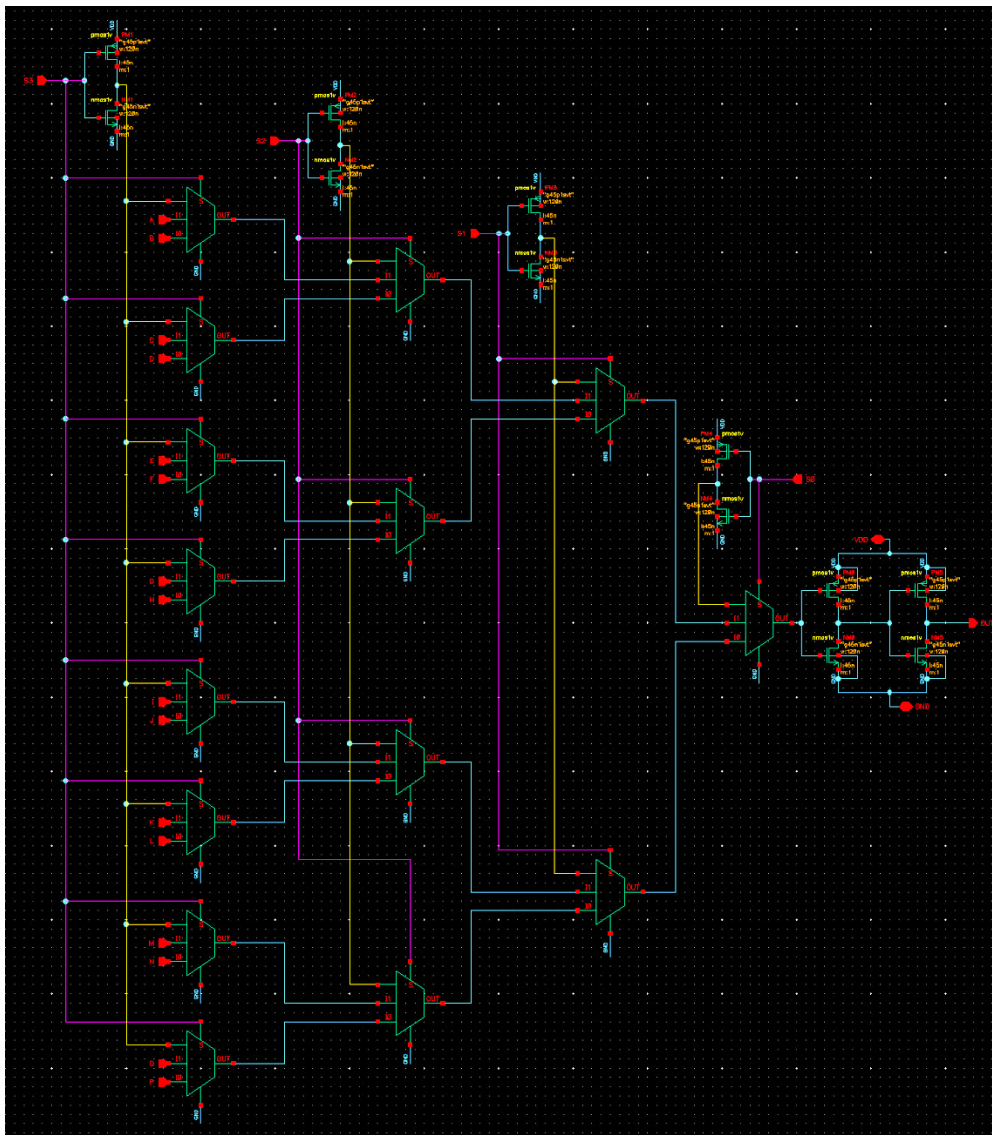


Figure 30: 16X1 LUT Schematic

LUT Symbol

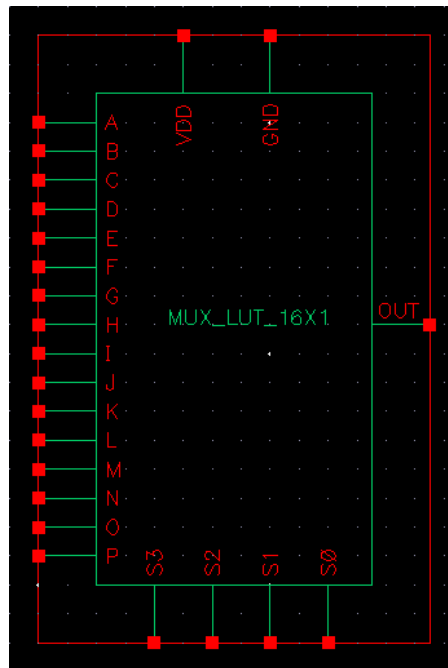


Figure 31: 16X1 LUT Symbol

Verification

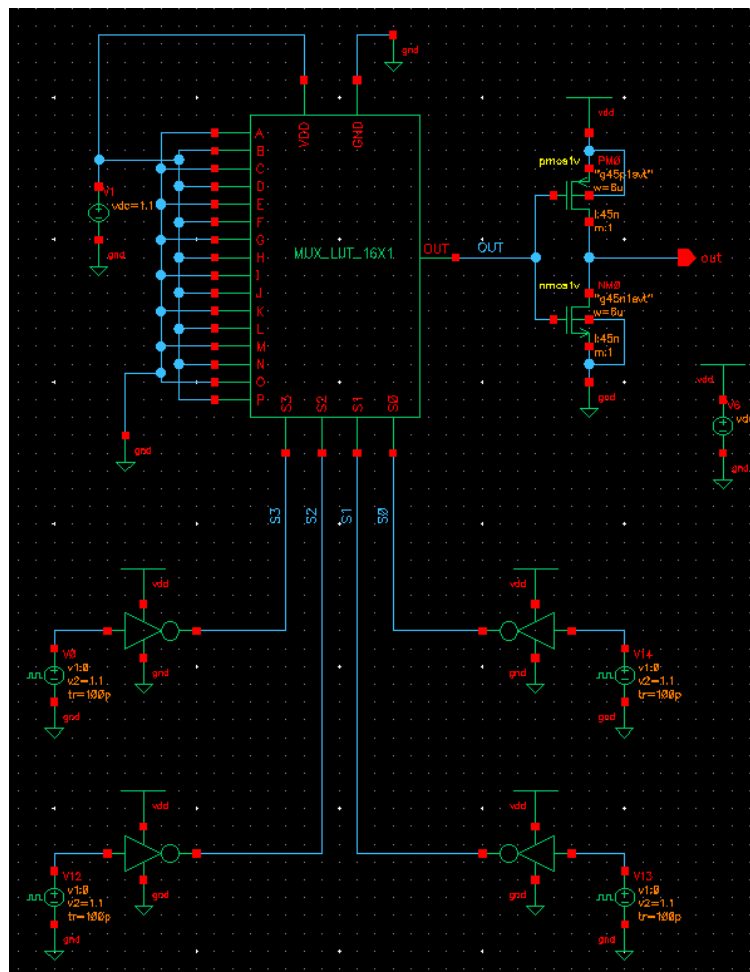


Figure 32: Test Schematic for 16X1 LUT

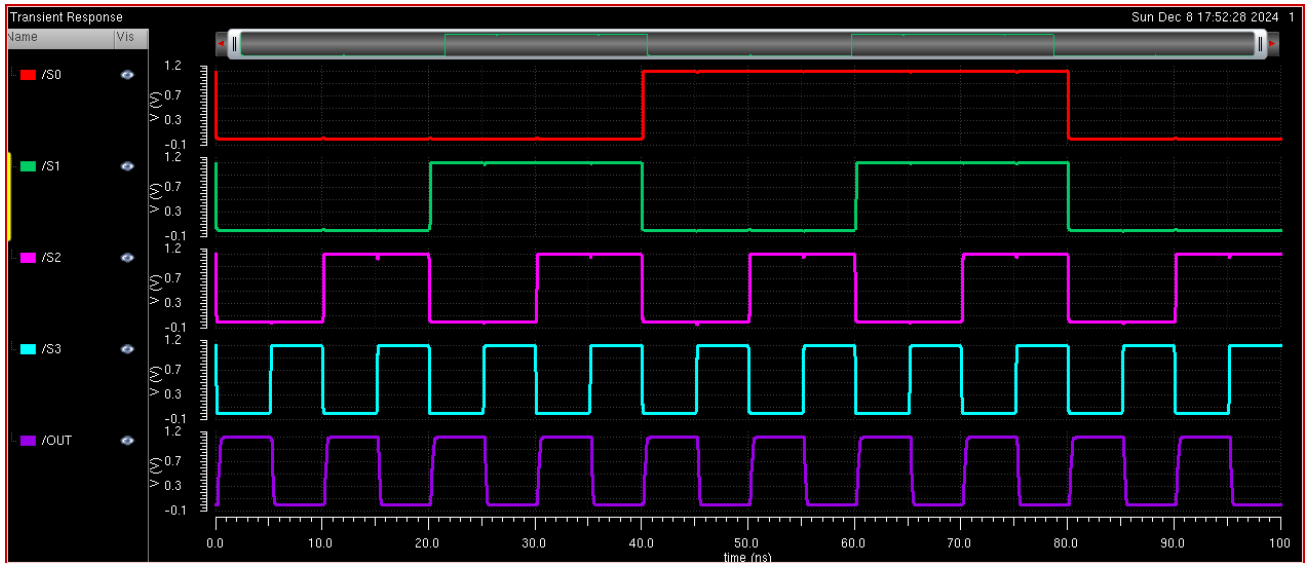


Figure 33: Verification Result for 16X1 LUT

LOAD LOGIC

The SIPO's load logic controls when the clock runs to transfer data. When load is low, the clock operates and shifts data into the SIPO. When load is high, the clock stops, preventing new data transfer. It is designed using 2 transmission gates to control both CLK and CLK_BAR.

Schematic

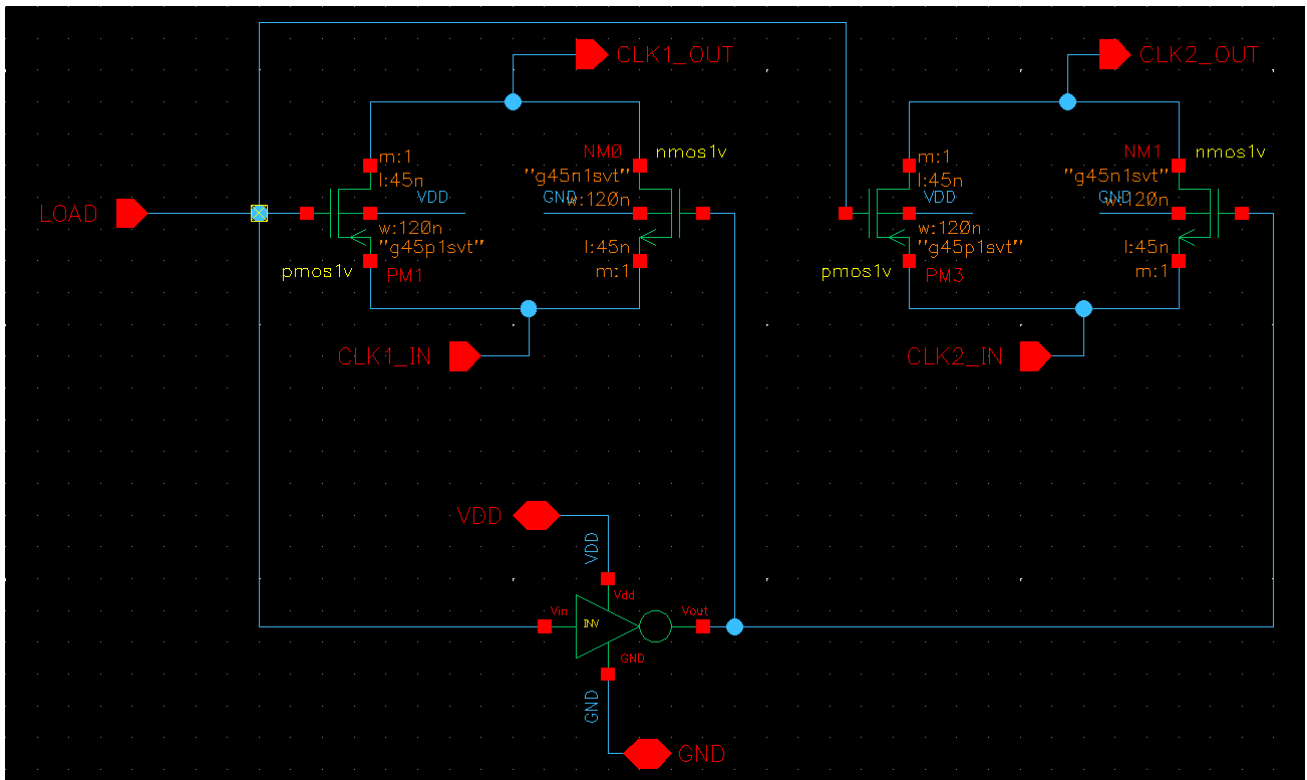


Figure 34: Load Logic Schematic

Symbol

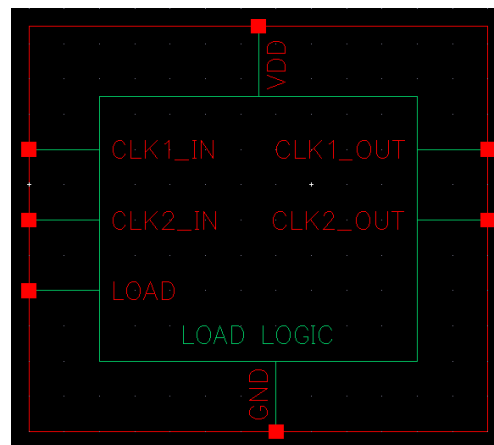


Figure 35: Load Logic Symbol

SIPO-SRAM Verification

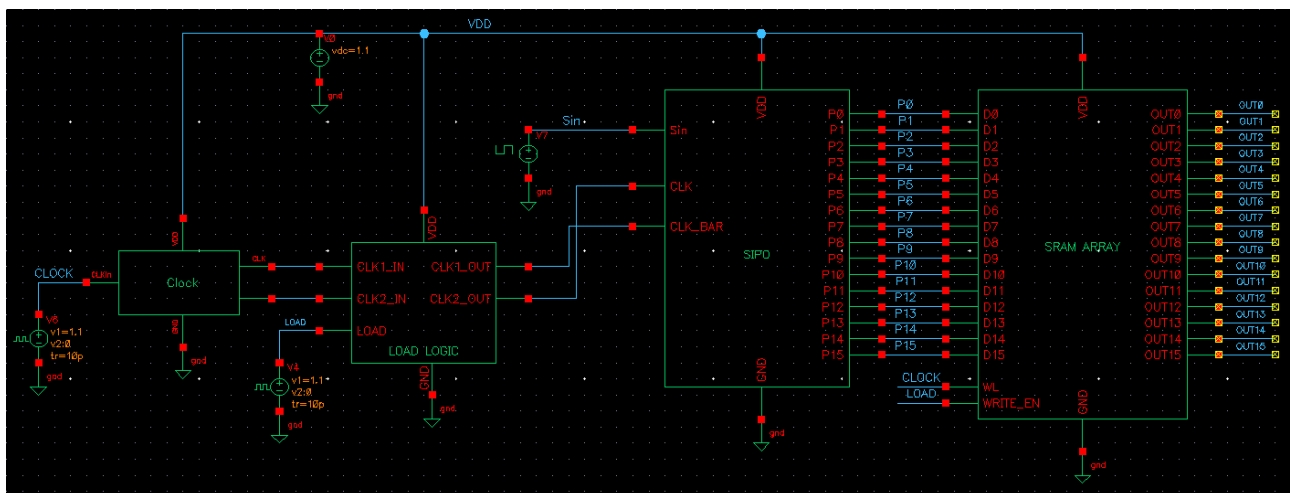


Figure 36: SIPO - SRAM Test Schematic

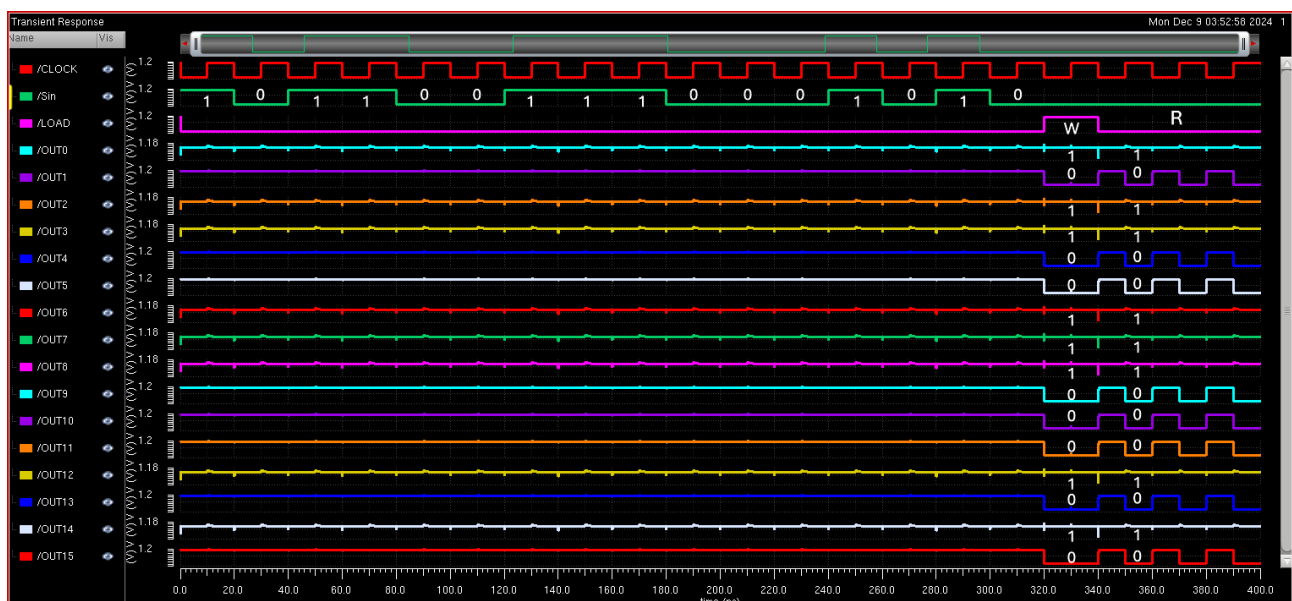


Figure 37: Verification Result for SIPO – SRAM

With reference to Fig. 36 and Fig. 37, the schematic and corresponding waveform validates the proper transmission of data from the SIPO to the SRAM and tests the accuracy of both write and read operations at the SRAM level.

- **Initial Data Loading into SIPO:**

As observed from the waveform, it takes 16 clock cycles to load data into the 16 D flip-flops of the SIPO register. During this time, the **Load** signal is kept **LOW**, allowing the SIPO clock to function and ensuring correct data shifting and transmission.

- **Transition to SRAM Write Operation:**

At the 17th clock cycle, the **Load** signal is set to **HIGH**. This action halts the SIPO clock, preventing further data shifting and stabilizing the output of the SIPO register.

Simultaneously, the SRAM begins the write operation, transferring the data from the SIPO outputs into the SRAM cell array.

The SRAM performs the write operation in a single clock cycle, as the data is written to all cells in parallel.

- **Read Operation from SRAM:**

After completing the write operation, the **Load** signal is reset to **LOW**, enabling the SRAM to perform the read operation. From the waveform, it is evident that the SRAM successfully writes and reads back the data, confirming the accuracy of the operations.

This validation confirms the seamless transfer of data from the SIPO to the SRAM and verifies the correct functionality of the SRAM write and read mechanisms.

SIPO-SRAM-LUT Verification

The below schematic and corresponding waveform validates the proper transmission of data from the SRAM to the LUT.

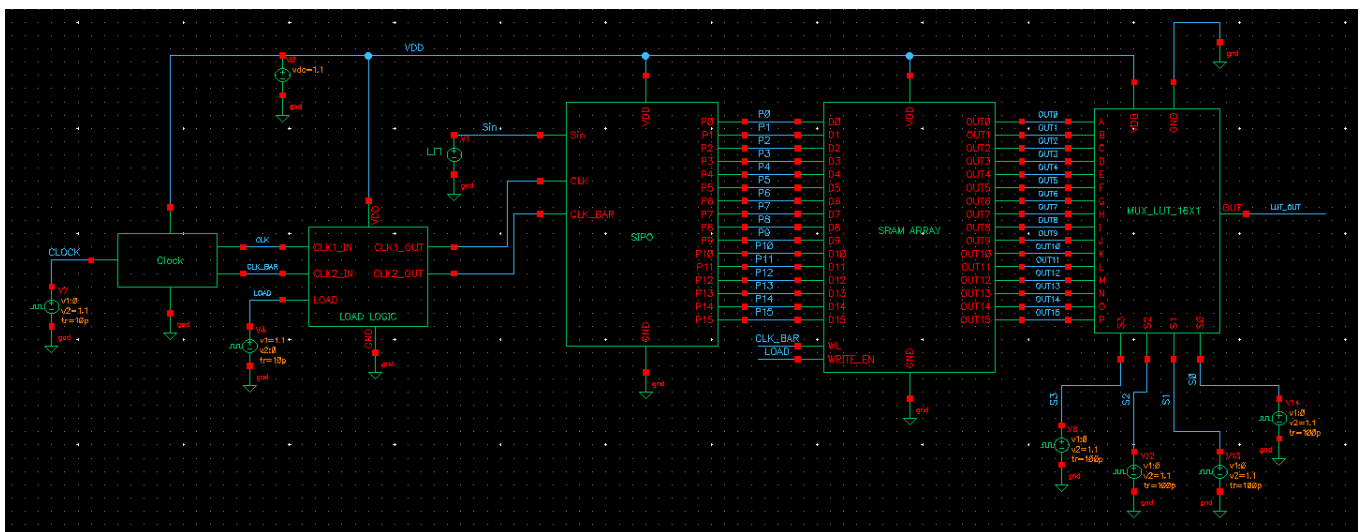


Figure 38: SIPO-SRAM-LUT Schematic

To implement the desired functionality, the Look-Up Table (LUT) in the Configurable Logic Block (CLB) is designed to produce specific outputs based on the input values provided through the select lines. Each select line combination corresponds to a predefined logic operation or value stored within the LUT. For this LUT, the output behavior can be described as -

S0	S1	S2	S3	OUT	
0	0	0	0	P	0
0	0	0	1	O	1
0	0	1	0	N	0
0	0	1	1	M	1
0	1	0	0	L	0
0	1	0	1	K	0
0	1	1	0	J	0
0	1	1	1	I	1
1	0	0	0	H	1
1	0	0	1	J	1
1	0	1	0	F	0
1	0	1	1	E	0
1	1	0	0	D	1
1	1	0	1	C	1
1	1	1	0	B	0
1	1	1	1	A	1

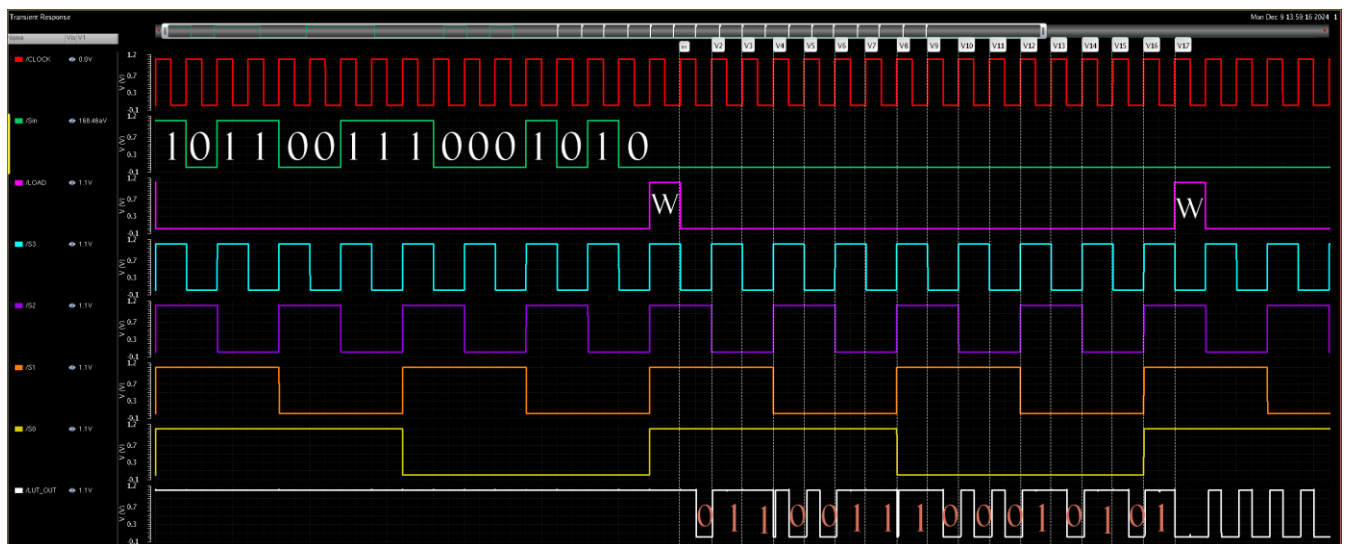


Figure 39: Verification Result for SIPO - SRAM – LUT

The graph demonstrates that the LUT is correctly outputting data corresponding to the inputs from the select lines. However, the output is not entirely smooth, as precharging occurs before the read data is stabilized. This precharging introduces a brief period of instability in the output signal.

To address this issue and achieve a smooth output without precharge artifacts, a **D flip-flop** is employed at the end of the LUT. The D flip-flop holds the stable output data for the duration of the clock cycle, effectively skipping the precharge phase. By capturing and storing only the valid read data, the D flip-flop ensures a consistent and reliable output signal during each clock cycle.

This approach eliminates transient fluctuations caused by precharging and enhances the overall stability of the CLB's output.

Schematic

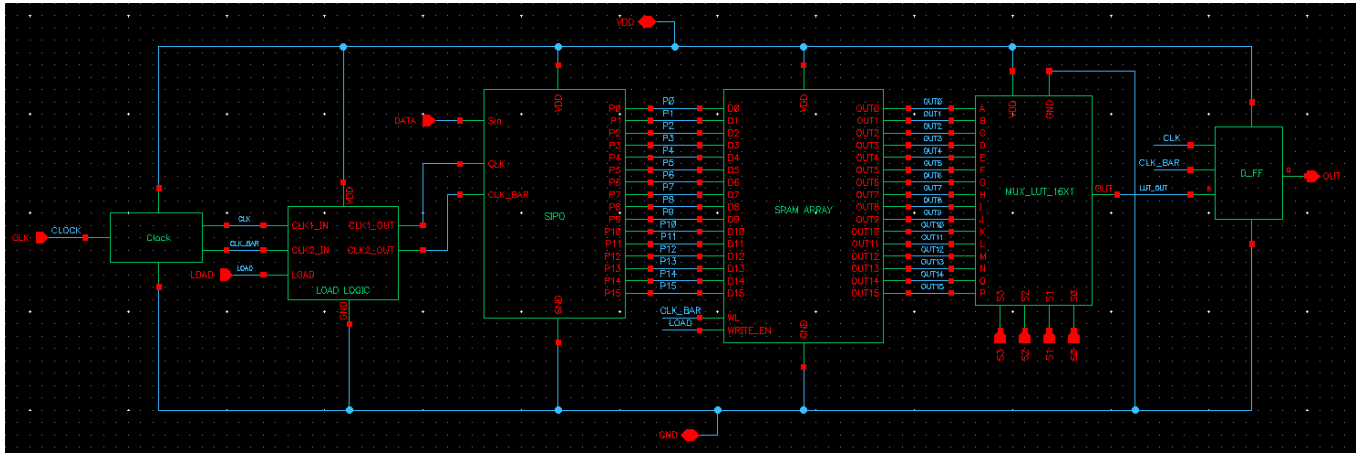


Figure 40: CLB schematic

Symbol

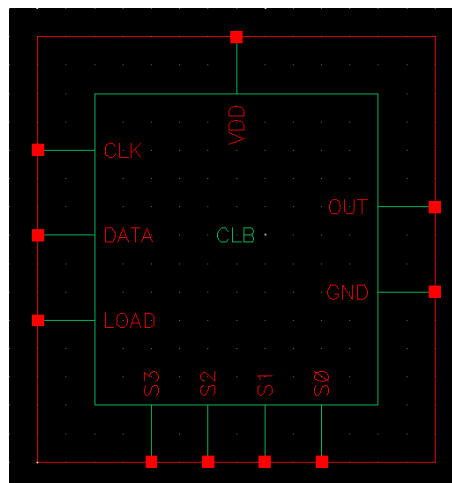


Figure 41: CLB Symbol

To ensure the proper functioning of the CLB, we must synchronize the clock time period for all the components.

	Time Period	Pulse width
Main clock	t	t/2
Data (Using Vbit)	t	-
Load	(16 * t) + t	16 * t
S3	period	period /2
S2	2 * period	period
S1	4 * period	2 * period
S0	8 * period	4 * period

Where period = $2 * t$

Verification

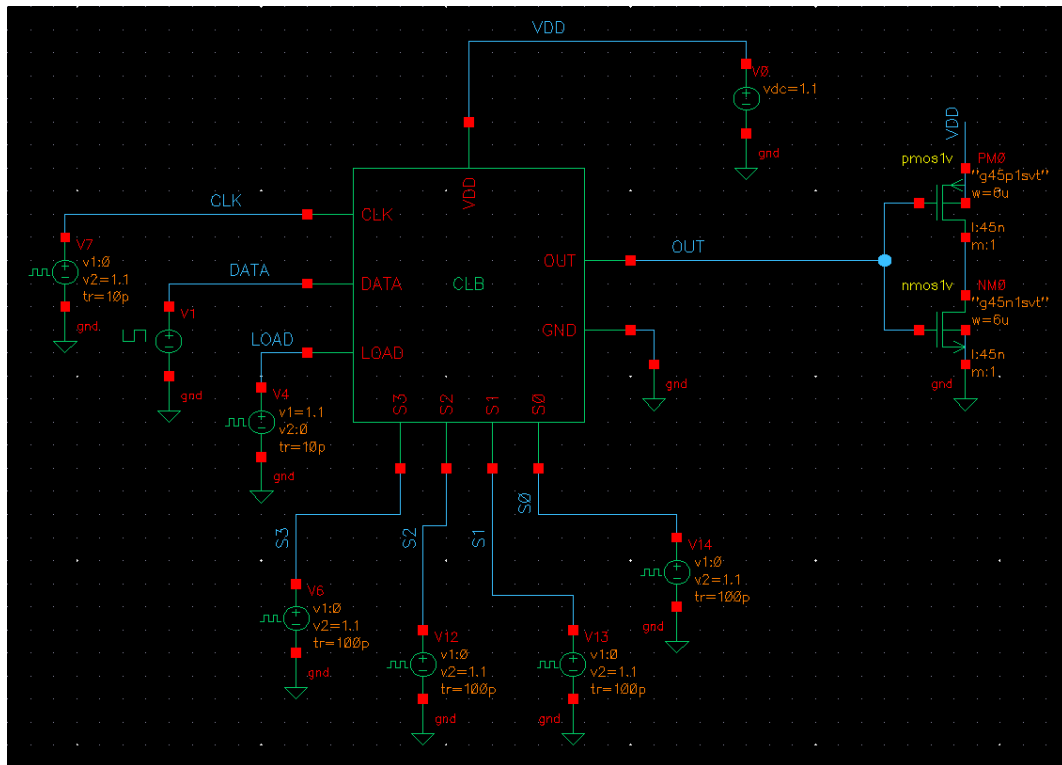


Figure 42: CLB Test Schematic

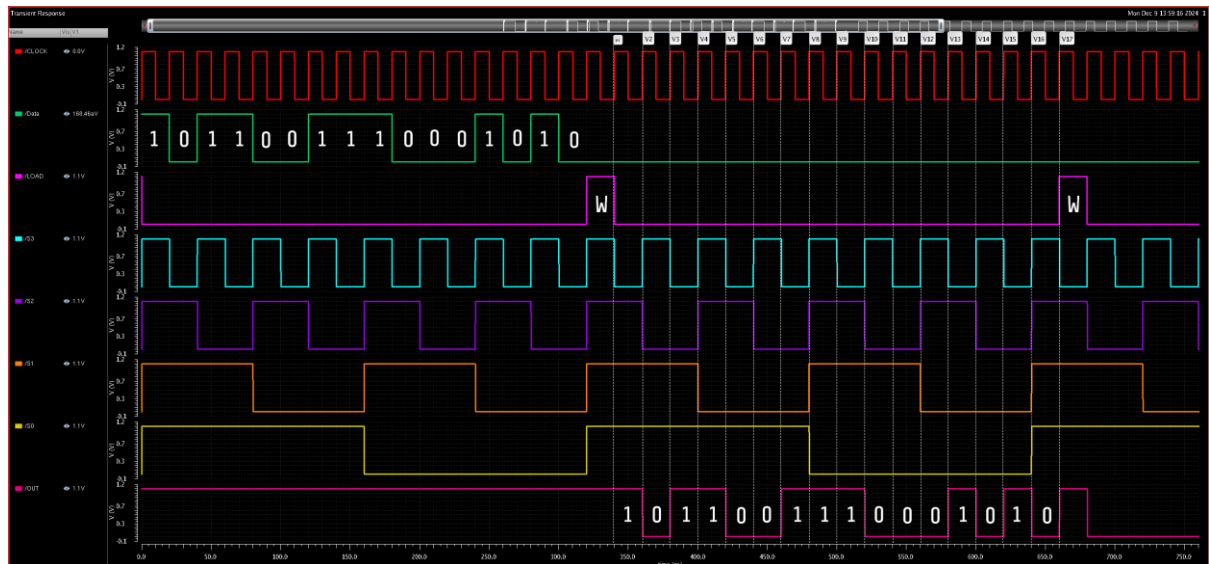


Figure 43: Verification Result for CLB

Optimisation Steps Taken

1. **SRAM Cell Sizing:**

The SRAM cells were carefully sized to ensure correct and reliable operation while maintaining efficiency in terms of power and speed.

2. **Centralized Prescaling Logic:**

Instead of including the prescaling logic within each SRAM cell, it was implemented as a common logic shared by the entire SRAM array. This optimization significantly reduced the overall cell area, improving layout efficiency.

3. **Common Inverter for LUT:**

The inverter from each 2x1 MUX in the 16x1 LUT was relocated to function as a common source for the entire LUT. This reduced redundancy, minimized area, and streamlined the design.

4. **Sized NAND Gates in Clock Generator:**

The NAND gates used in the clock generator were optimized by resizing them for faster operation, ensuring improved timing performance and reduced delays in clock signal generation.

Frequency

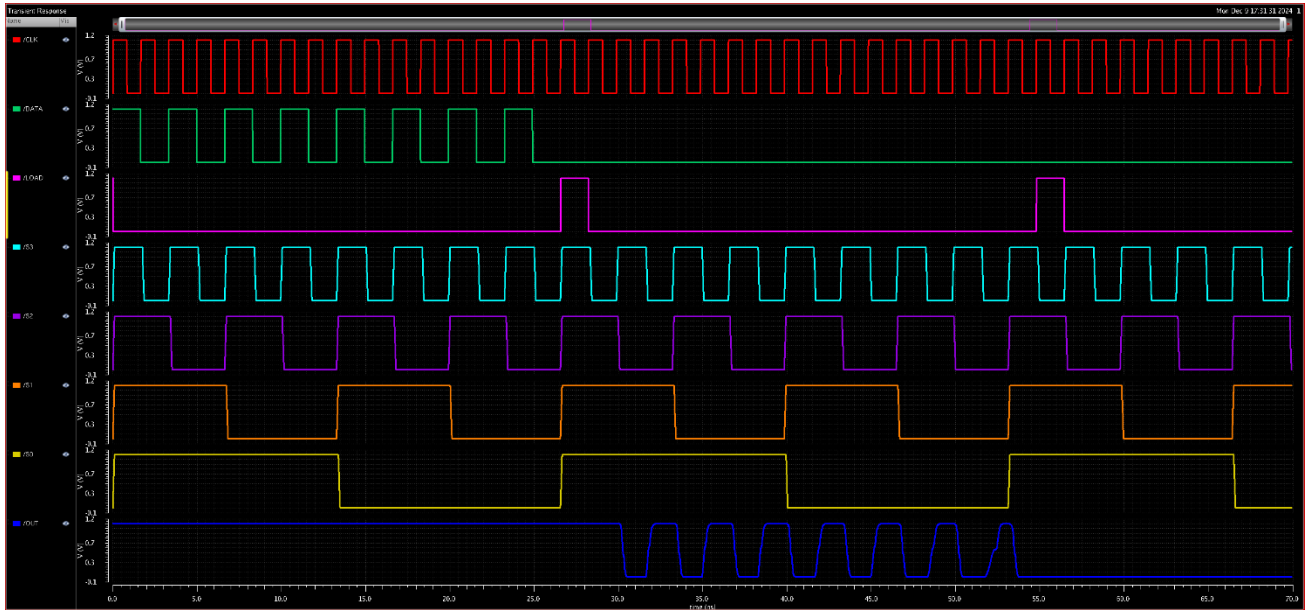


Figure 44: Max Frequency giving correct output at 1.66ns time period

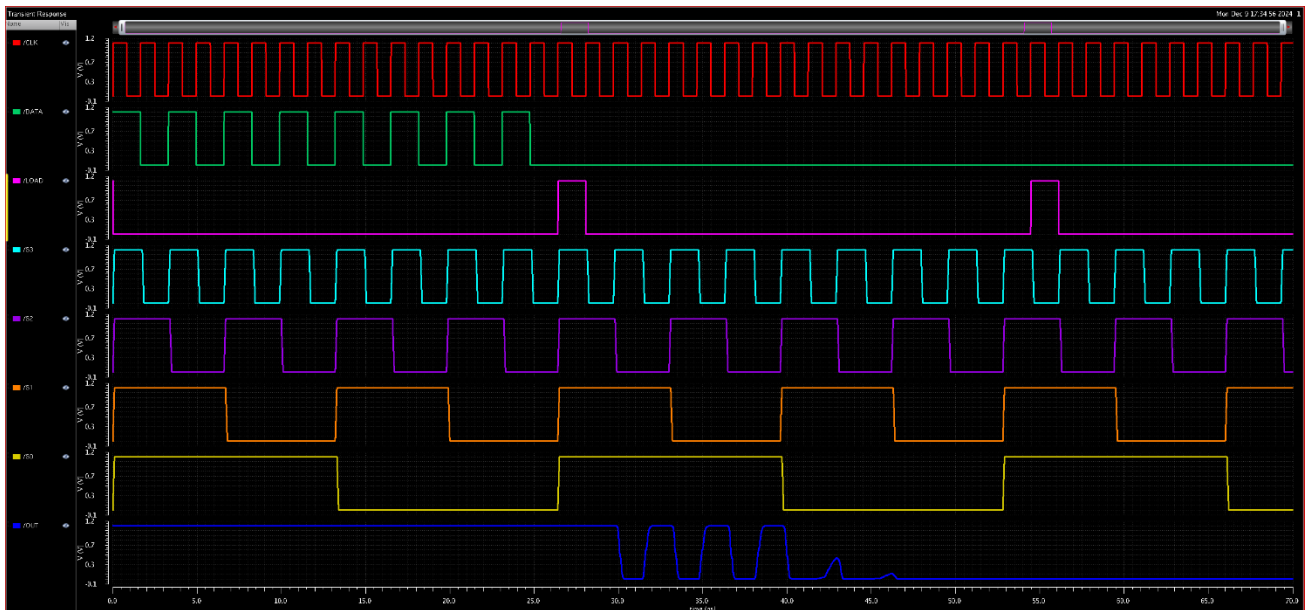


Figure 45: Incorrect Output at 1.65ns time period

This test case is chosen as this will provide us with the worst case switching scenario.

1010101010101010

From the above 2 graphs we can conclude that the max switching frequency at which the CLB operates correctly is –

$$\text{Max Frequency} = \frac{1}{\text{Time Period}} = \frac{1}{1.66\text{ns}} = \mathbf{0.6\ GHz}$$

Average Energy

Loading Energy

For the loading energy measurement, we know that the data first gets collected in SIPO for 16 clock cycles and pushed to SRAM array parallelly in 1 clock cycle.

Hence, we will measure the energy for 17 clock cycle.

Since the operating period of our design is 1.66 ns,

The integration is done for 0 to $1.66 * 17 = 28$ ns. i.e **(0 – 28ns)**

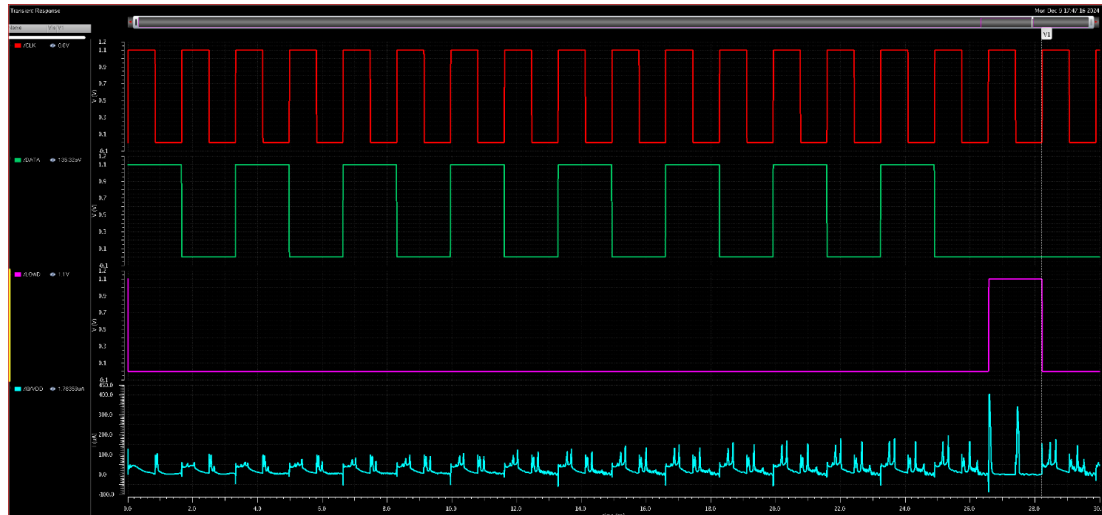


Figure 46: Loading Energy from 0 to 28ns

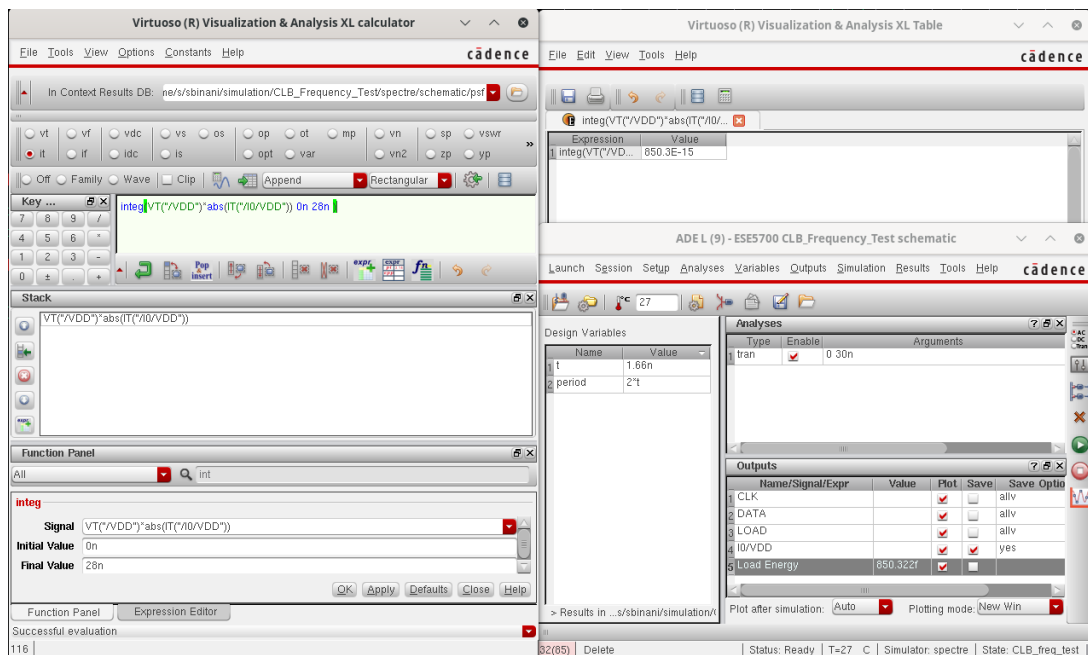


Figure 47: Loading Energy Calculation

Calculated Loading Energy = **850.32 fJ**

Active Energy

Similar to the calculation of active energy, it takes 16 more clock cycles to cycle through all address bits from 0 to 15 i.e $1.66 * 16 = 26.56$ ns

Hence, the integration is done for 28 to (28 + 26.56) i.e **(28 - 54)ns**

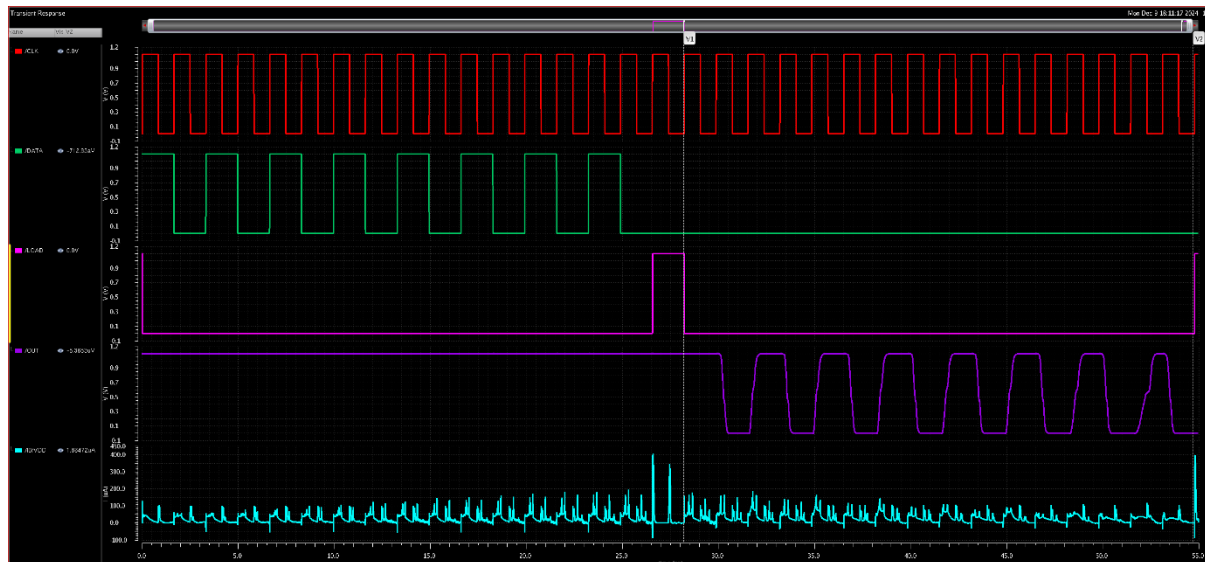


Figure 48: Active Energy from 28ns to 54ns

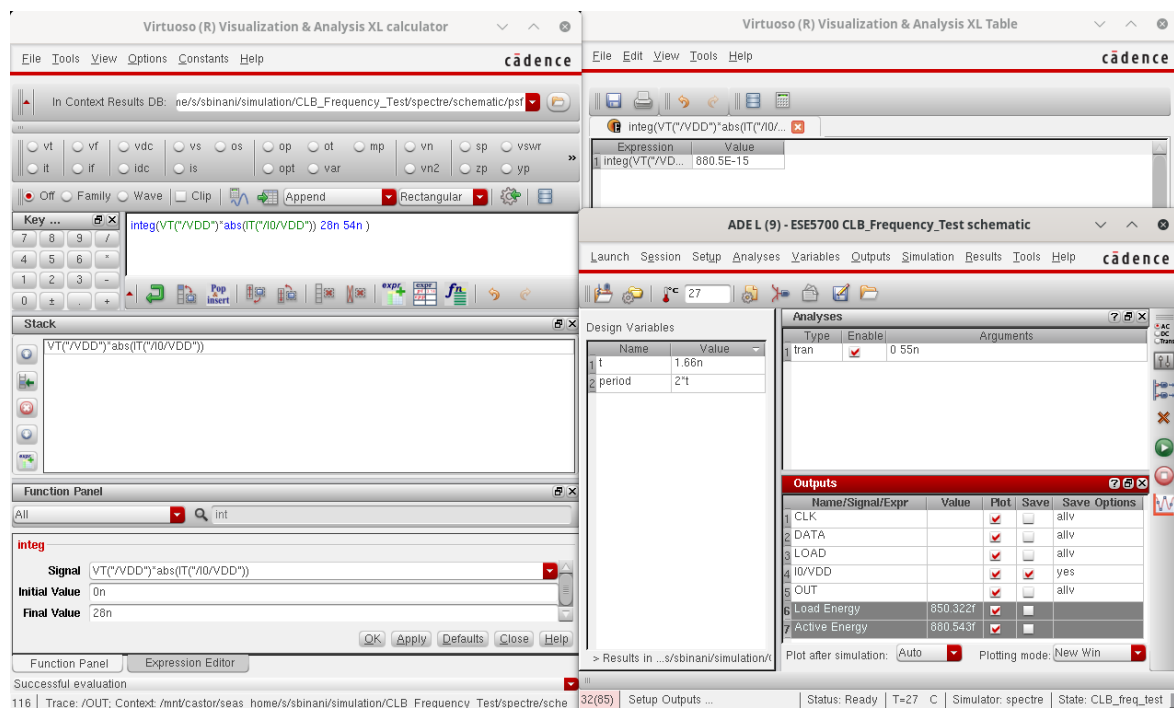


Figure 49: Active Energy Calculation

Calculated Active Energy = **880.54 fJ**

$$\begin{aligned}
 \text{Average Energy} &= 20\% \text{ Loading Energy} + 80\% \text{ Active Energy} \\
 &= (0.2 * 850.32) + (0.8 * 880.54) \text{ fJ} \\
 &= \mathbf{874.496 \text{ fJ}}
 \end{aligned}$$

Area

Cell Area

$$\begin{aligned}
 \text{Area}_{\text{cell}} &= W_{\text{precharge}} + W_{\text{data_inverter}} + W_{\text{Write_enable}} + W_{\text{access}} + W_{\text{cross_coupled_inv}} \\
 &= (2 * 120) + (2 * 120) + (2 * 240) + (2 * 120) + ((2 * 120) + (2 * 180)) \text{ nm} \\
 &= \mathbf{1800 \text{ nm}}
 \end{aligned}$$

2X1 Mux Area

$$\begin{aligned}
 \text{Area}_{\text{mux}} &= 2 * 120 \text{ nm} \\
 &= \mathbf{240 \text{ nm}}
 \end{aligned}$$

$$\begin{aligned}
 \text{Total area} &= 16 * \text{Area}_{\text{cell}} + 15 * \text{Area}_{\text{mux}} \\
 &= (16 * 1800) + (15 * 240) \text{ nm} \\
 &= \mathbf{32400 \text{ nm}}
 \end{aligned}$$

$$\begin{aligned}
 \text{FOM} &= \text{Area} * \text{Average Energy} * \frac{1}{\text{Max Frequency}} \\
 &= 32400 \text{ nm} * 874.496 \text{ fJ} * 1.66 \text{ ns} \\
 &= \mathbf{47 \times 10^{-26} \text{ mJs}}
 \end{aligned}$$

Summary Table

Design Metrics	Value
Max Frequency	0.6 GHZ (1.66 ns)
Average Energy	874.496 fJ
Area	32400 nm